



**Verified Carbon
Standard**

IACO REDD+ PROJECT



IACO REDD+

Document Prepared By WayCarbon Soluções Ambientais e Projetos de Carbono LTDA

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1 PROJECT DETAILS

1.1 Summary Description of the Project

In the last three years (2018-2021), 111.6 hectares were deforested per hour in the Brazilian Amazon biome, which equates to about 18 trees per second. Agriculture and livestock were responsible for almost all of the country's deforestation, being the cause for more than 97%¹ of the cases. The IACO REDD+ Project seeks to address the issue of deforestation in the Amazon biome on a local level at the state of Acre, Brazil. The IACO REDD+ Project is an AFOLU project, and therefore is classified within scope 14. More specifically, the project falls under the Avoided Unplanned Deforestation (AUD) and Avoided Planned Deforestation (APD) categories.

Sena Madureira, the municipality where the project is located, has the second largest area of accumulated deforestation in the Acre state for the period between 1988 and 2020, summing up to 212,917.6 hectares². In 2020, the municipality also had the second largest deforested area in the state, of 7,322.2 hectares, which represents 11.8% of the deforested area in Acre². Also, the project area is surrounded by conservation units (UCs) that had the highest deforestation rates in the state in recent years³. Much of this deforestation is driven by the expansion of livestock and soy cultivation, logging, growing infrastructure, and colonization by subsistence farmers. Particularity in the IACO REDD+ project region, areas of native forest have been cleared mainly for livestock and subsistence colonization. With the increasing pressure for deforestation, the original Amazon rainforest present in the IACO REDD+ project is threatened and the lack of economic return from the property is the major barrier to the implementation of activities that prevent the conversion of forest to other land uses. If this REDD+ project is implemented, 147,398 hectares of Amazon rainforest will be conserved over the first 30 years, avoiding the emission of 21,043,825.57 tons of CO₂ into the atmosphere. A significant part of these reduced emissions will be achieved through the abandonment of the planned deforestation that was about to occur in 20% of the property's area.

Through carbon revenue, IACO's REDD+ proponents intend to install sign boards to inform the presence of the REDD+ project, hire forest patrols, build the project's base camp, provide courses and training for fire prevention and fighting, and provide means for community awareness around the project about climate change, biodiversity and ecosystem services, and sustainable management of forests, agricultural land and grasslands is improved. In addition, high-resolution satellite images will be acquired to monitor forest cover and biodiversity conservation. With these activities, the project will directly and indirectly contribute to the mitigation of global climate change and the reduction of occurrences and risks associated with extreme weather events.

Under the social sphere, the project also brings benefits for the region. The IACO REDD+ project proponents recognize the essential role of the participation of project area residents and neighbors in the success of this project. Then, the improvement of the quality of life and environmental conscience will be worked throughout the duration of the project and is expected to prevail after its crediting period.

¹ Relatório Anual de Desmatamento 2021 - São Paulo, Brasil. MapBiomias, 2022 - 126 páginas. <http://alerta.mapbiomas.org>

² Dinâmica do desmatamento no Estado do Acre – PRODES ano florestal 2019/2020. GOVERNO DO ESTADO DO ACRE, 2021

³ Zoneamento ecológico-econômico do Acre: fase III: escala 1:250.000: documento-síntese / Secretária de Estado do Meio Ambiente. – Rio Branco: Semapi, 2021

Activities focused on improvement and diversification of the local economy, education and local health are planned and described on section 1.11.

1.2 Sectoral Scope and Project Type

The sectoral scope applied to the IACO REDD+ Carbon Project is scope 14 - Afforestation, Forestation and Other Land Use (AFOLU), specifically under the Reducing Emissions from Deforestation and Degradation (REDD) through Avoided Unplanned Deforestation (AUD) and Avoided Planned Deforestation (APD). This is not a grouped project.

1.3 Project Eligibility

The IACO REDD+ Project meets all the eligibility criteria and requirements set by the VCS Program Version 4.3 for the AFOLU sectorial scope.

- The IACO REDD+ Project applies a methodology eligible under the VCS Program, VM0007.
- The project is located in the state of Acre, which intends to incorporate incentives and projects related to the reduction of emissions from deforestation and forest degradation – REDD+ program zones. So far, as further described under section 1.14, the Acre REDD+ Jurisdictional Program is a document without force of law and was not formally validated under the VCS, or any other standard of REDD+ in Brazil or worldwide. Currently, this program is just a guidance document for the public administration, which is applicable to the State's action in reducing GHG emissions from deforestation.
- The implementation of the IACO REDD+ Project activities do not lead to the violation of any applicable law;
- The IACO REDD+ Project does not convert native ecosystems to generate GHG. In fact, the project aims to avoid the native ecosystem conversion. Nevertheless, there has not been a change in land use in the project activity area within the last 10 years before the project start date. Images below show the project activity areas 10 years before the project start date and in 2020.

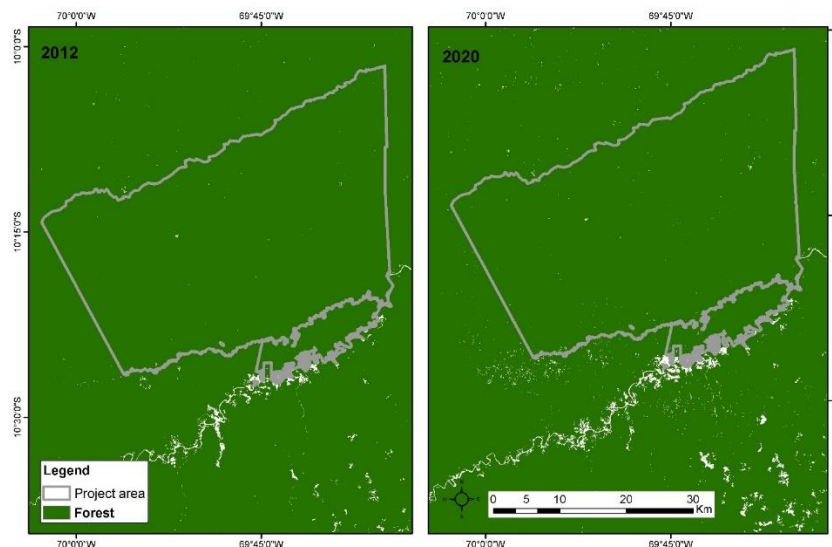


Figure 1 – Forest cover of IACO REDD+ Project area in 2012 and 2020. Fonte: MapBiomass Collection 6.0⁴

- The IACO REDD+ Project does not occur on wetlands and does not drain native ecosystems or degrade hydrological functions; In addition, the Permanent Preservation Areas (APP), including the areas covered by native vegetation that border the rivers and are part of the project area, are preserved.
- The Non-permanence risk will be analyzed by the VCS Program document AFOLU Non-Permanence Risk Tool version 4.0.

1.4 Project Design

The IACO REDD+ project has been designed as an Avoided Planned and Unplanned Deforestation applying VM0007 methodology, Version 1.6, 08 September 2020, in a single location (see Figure X).

Eligibility Criteria

Not applicable. This is not a grouped project.

1.5 Project Proponent

Organization name	IACO Ambiental LTDA.
Contact person	Rodrigo Gomes De Oliveira and Diego Hoebel Munhoz
Title	Partners of IACO Ambiental LTDA.
Address	890, Espírito Santo Street, ZIP-Code: 86.010-510, Londrina city, Paraná, Brazil
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Organization name	WAYCARBON SOLUÇÕES AMBIENTAIS E PROJETOS DE CARBONO LTDA
Contact person	Felipe Bittencourt
Title	Chief Executive Officer
Address	1000, Paraíba Street, 7th floor. Belo Horizonte/MG. Brazil.

⁴ MapBiomass Project, <https://mapbiomas.org/produtos>

Telephone	+55 31 3656-0501
Email	fbittencourt@waycarbon.com

1.6 Other Entities Involved in the Project

No other entities are involved in the project.

1.7 Ownership

The project is conducted within a private property, named “Fazenda Guanabara/Petrópolis”, which has 148,541.15 hectares and is located in the municipality of Sena Madureira, in the state of Acre, Brazil. This property is composed of 68 parcels (designated by “Local ID” on **Table 1 – Farm name, Property deeds, Local ID, CAR-Code**) with CAR registrations that were leased by the IACO Ambiental Ltda.

The information of the parcels that compose the project area is in the table below. Property deeds and their CAR registry numbers⁵ are listed for each respective local ID (i.e. local name of parcel). **Figure 2** presents the delimitation and arrangement within the project area.

Table 1 – Farm name, Property deeds, Local ID, CAR-Code

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Guanabara	Lote 1	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 30-N; Page.073	438 Book 2RG	AC-1200500-3B8DE590124C4AFCB86D407144E6CF82
Guanabara	Lote 2	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 29-N; Page.095	446 Book 2RG	AC-1200500-FC24D2FD589041C285ED92A36257A47E

⁵ Cadastro Ambiental Rural (CAR), <https://www.car.gov.br/#/>

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Guanabara	Lote 3	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 29-N; Page.097	445 Book 2RG	AC-1200500-A92A0DCBD6584AC982A9395CBF465BB5
Guanabara	Lote 4	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 158; Page.068	436 Book 2RG	AC-1200500-3AD551D8ABA949CB9B1CC898E27B567D
Guanabara	Lote 5	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 30-N; Page.074	437 Book 2RG	AC-1200500-E0C9D24C45484E13916726F9EB06687E
Guanabara	Lote 6	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 29-N; Page.104	444 Book 2RG	AC-1200500-8D1B1D0A2CF64E3680FFA66BB7C27005
Guanabara	Lote 7	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of	443 Book 2RG	AC-1200500-894912D07E6E49E6BEA4E56B3CC3658D

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
		Paraná; book number 60-N; Page.049		
Guanabara	Lote 8	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 29-N; Page.103	448 Book 2RG	AC-1200500-33EE63B7F91E4D8DBFD7C5C19AD5E195
Guanabara	Lote 9	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 29-N; Page.096	449 Book 2RG	AC-1200500-2DC921C828F2441BAC108ECB11C06
Guanabara	Lote 10	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 149; Page.184	330 Book 2	AC-1200500-75F30AC0D32746F0AB4C9E33167B33C9
Guanabara	Lote 11	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 135; Pages. 29 and 30	331 Book 2RG	AC-1200500-CCE71955CC7E4095ACDCF1F4D8D65CF0

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Guanabara	Lote 12	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 166; Page.126	364 Book 2RG	AC-1200500-374BA7C200D5445788 E865215755B7E9
Guanabara	Lote 13	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 129; Page.155 and 156	363 Book 2RG	AC-1200500-C2ADDB8319204E139AE D53540CAAF9C6
Guanabara	Lote 14	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 60-N; Page.043	450 Book 2RG	AC-1200500-F97D79A2F5004F04852 2228BB1A5FBF6
Guanabara	Lote 15	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Rio Branco Registry, from the municipality of Rio Branco, state of Acre; book number 2-A; Page.273	451 Book 2G	AC-1200500-507E36B44B1349B3A18 AAF5526BA7805
Guanabara	Lote 16	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of	332 Book 2RG	AC-1200500-2CD876EB495E44CDB8 7E30E68451CF22

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
		Faxinal, state of Paraná; book number 113; Page.182		
Guanabara	Lote 17	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cândido de Abreu Registry, from the municipality of Cândido de Abreu, state of Paraná; book number 047; Pages. 017 to 019.	333 Book 2RG	AC-1200500-DAD78316048A4500B064DEE0C1A8F2D7
Guanabara	Lote 18	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 167; Page.141	320 Book 2RG	AC-1200500-5C6F29D1F4E44B689B8DD75DC4F364DF
Guanabara	Lote 19	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 167; Page.139	307 Book 2RG	AC-1200500-01059E84CA084A9FA45D7FF164BDDD24
Guanabara	Lote 20	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 167; Page.140	325 Book 2RG	AC-1200500-AB441DDA11BA41D7865C1899EA4DE6A7

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Guanabara	Lote 21	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 167; Page.142	334 Book 2RG	AC-1200500-2DECDF6F1934433E91A B89268372B43D
Guanabara	Lote 22	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 167; Page.143	326 Book 2RG	AC-1200500-C225FAD69EA148CDBD9 32F601AD99D4A
Guanabara	Lote 23	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 167; Page.144	327 Book 2RG	AC-1200500-D5F9A448B88D47A3BC D2B4E9EC37C759
Guanabara	Lote 24	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 167; Page.134	328 Book 2RG	AC-1200500-BF7F3EDE2EA048EDB2B 8F41932FC5A1D
Guanabara	Lote 25	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of	321 Book 2RG	AC-1200500-0162310082EA4DF187C 545D41A8A229C

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
		Paraná; book number 167; Page.135		
Guanabara	Lote 26	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 167; Page.137	322 Book 2RG	AC-1200500-EA33A111BF07408989A9CF86B2868EC1
Guanabara	Lote 27	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 167; Page.136	323 Book 2RG	AC-1200500-CF7963844F364308BF8CCC2398A88A6B
Guanabara	Lote 30	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 60-N; Page.041	313 Book 2RG	AC-1200500-72C4094579844D87A3E0F8F15CD6EDC1
Guanabara	Lote 31	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 29-N; Page.098	441 Book 2RG	AC-1200500-E0FFACA05257439BB2DB868C02CF1AAD
Guanabara	Lote 32	PUBLIC WRITING OF PURCHASE AND SALE, registered in the	556 Book 2RG	AC-1200500-BDDEED483F974A95AB3E2D880B45 ACE1

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
		Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 120; Pages.125 and 126		
Guanabara	Lote 33	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 153; Page.093	319 Book 2RG	AC-1200500-45FA30CB13114E669140DC7768A0F0E4
Guanabara	Lote 34	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 153; Page.092	329 Book 2RG	AC-1200500-F052643AB36E465AA817A30403A15986
Guanabara	Lote 35	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 163; Page.002	324 Book 2RG	AC-1200500-68B9EC8D7DBA41DC9BF1D7387961E1EE
Guanabara	Lote 36	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Londrina Registry, from the municipality of Londrina, state of Paraná; book number 60-N; Page.047	447 Book 2RG	AC-1200500-5BE51166825D471E99BDF2D06F0595CF

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Guanabara	Lote 37	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 60–N; Page.050	442 Book 2RG	AC-1200500-5CFF1002338B4F3EB0EFC2562C9E4D48
Guanabara	Lote 38	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 60–N; Page.046	465 Book 2A	AC-1200500-F6C3660510B8427B91EE8D02122C6C0A
Guanabara	Lote 39	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 60–N; Page.044	471 Book 2A	AC-1200500-F4D1AC43F76D4B579E21DEFB7D5D1FD5
Guanabara	Lote 40	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 171; Pages.181 and 182	340 Book 2A	AC-1200500-EA5A585CD28341A2B8BECBCAE288CFCF
Guanabara	Lote 41	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of	341 Book 2RG	AC-1200500-71108205CFA947B48ABBDDF060B457B

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
		Faxinal, state of Paraná; book number 153; Page.003		
Guanabara	Lote 42	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 153; Page.005	342 Book 2RG	AC-1200500-88D88A437FAF4A91B44 ABB74B8E2BB1D
Guanabara	Lote 43	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 60-N; Page.048	470 Book 2RG	AC-1200500-CCB1C1D3289A414Aafb D58ECBB67C3F2
Guanabara	Lote 44	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Londrina Registry, from the municipality of Londrina, state of Paraná; book number 60-N; Page.048	464 Book 2RG	AC-1200500-F58856D181BD4984BB E87F905B4510F9
Guanabara	Lote 46	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 166; Page.125	343 Book 2RG	AC-1200500-F585C0AFBFDF49DE9A7 B5AF5D891BCB4

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Guanabara	Lote 28-A	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Tangará da Serra Registry, from the municipality of Tangará da Serra, state of Mato Grosso; book number 038; Page.109	299 Book 2RG	AC-1200500-52286F5EF2D64DFA9EB91EC81612B25B
Guanabara	Lote 28-B	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Sena Madureira Registry, from the municipality of Sena Madureira, state of Acre; book number 001; Page.149	246 Book 2RG	AC-1200500-0883FF943A7542669247EF12D6F3FA04
Guanabara	Lote 32-A	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 60-N; Page.042	555 Book 2RG	AC-1200500-21E4B9F172D740E5A39C16BE1CEC70CA
Guanabara	Lote 35-A	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Cesário Registry, from the municipality of Londrina, state of Paraná; book number 60-N; Page.045	495 Book 2RG	AC-1200500-2EE242818C294CE9A5BD3EC23493C340

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Petrópolis	Lote 6	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 435-N; Page.097	1934 Book 2RG	AC-1200500-C7E80B9429404591BBA D37D1569EDE42
Petrópolis	Lote 8	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Rio Branco Registry, from the municipality of Rio Branco, state of Acre; book number 007; Pages.075 and 077	266 Book 2RG	AC-1200500-8FB034E4886B46B58B D966 AC3E28B20F
Petrópolis	Lote 9	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 434-N; Page.105	1933 Book 2RG	AC-1200500-AB1BEBF2BB564DCFB6F 73312AA94B93B
Petrópolis	Lote 11	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number ; Page.	2088 Book 2RG	AC-1200500-0EF9557381D8485E8F1 33E58AF836743

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Petrópolis	Lote 12	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Faxinal Registry, from the municipality of Faxinal, state of Paraná; book number 171; Pages.177 to 179	153 Book 2RG	AC-1200500-C476B49E60554C0F847394136922ABFB
Petrópolis	Lote 13	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 446-N; Page.107	1979 Book 2RG	AC-1200500-482108FD13AF4D1AAF5A225FA80BC376
Petrópolis	Lote 14	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 446-N; Page.046	1978 Book 2G	AC-1200500-E37521DBE49647F6BAF4A45616385EBF
Petrópolis	Lote 16	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 434-N; Page.117	17 Book 2RG	AC-1200500-164BB70D333348BCBABB95A7920BB627

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Petrópolis	Lote 13-A	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 447-N; Page.046	1977 Book 2G	AC-1200500-1BAFEF4D860047CBBB833B6689995B31
Petrópolis	Lote 15-A	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 434-N; Page.123	1935 Book 2F	AC-1200500-57D994DD239D4319B01383D68A1B6BDE
Petrópolis	Lote 15-B	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 434-N; Page.107	1940 Book 2F	AC-1200500-8066B865C95F455E8BF3C781AF310197
Petrópolis	Lote 15-C	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 434-N; Page.111	1941 Book 2RG	AC-1200500-BE94E1D9FA924DDFB72C4411421D153A

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Petrópolis	Lote 16-A	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 434-N; Page.115	1936 Book 2RG	AC-1200500-F439AABB775F4193A17C2E7589419690
Petrópolis	Lote 17-18	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 434-N; Page.113	1942 Book 2RG	AC-1200500-1A3345F6D6DD4B11AD5C52BE358167DE
Petrópolis	Lote 1-A	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Liana Claudia Registry, from the municipality of Maringá, state of Paraná; book number 434-N; Page.121	1943 Book 2	AC-1200500-756F1AE90A5F4496AD62CCF3519A59EE
Petrópolis	Lote 4-A	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 496-N; Page.051	1976 Book 2RG	AC-1200500-538B818CADD649778A39A4580DB67EF7

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Petrópolis	Lote 4-B	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 451-N; Page.023	2084 Book 2RG	AC-1200500-93AE98A4EF5046FAB36161812B4AD06B
Petrópolis	Lote 5	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 446-N; Page.061	1980 Book 2RG	AC-1200500-4810F4A40DE04F80AE886196D047FBD3
Petrópolis	Lote 6-A	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 435-N; Page.099	1938 Book 2RG	AC-1200500-64CBDA65B29C4812BE8FFF73920EB67A
Petrópolis	Lote 7-10	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 434-N; Page.109	1937 Book 2RG	AC-1200500-9B9871DF9E6E4C73822E9B9EDB2CCE13

Farm name	Local ID	Property deeds	Property Registry Office Number	CAR registry number
Petrópolis	Lote 9-A	PUBLIC WRITING OF PURCHASE AND SALE, registered in the Diogenes Pinto Registry, from the municipality of Maringá, state of Paraná; book number 486-N; Page.176	1916 Book 2RG	AC-1200500-557EE07948104C63B8636B30116B6A75

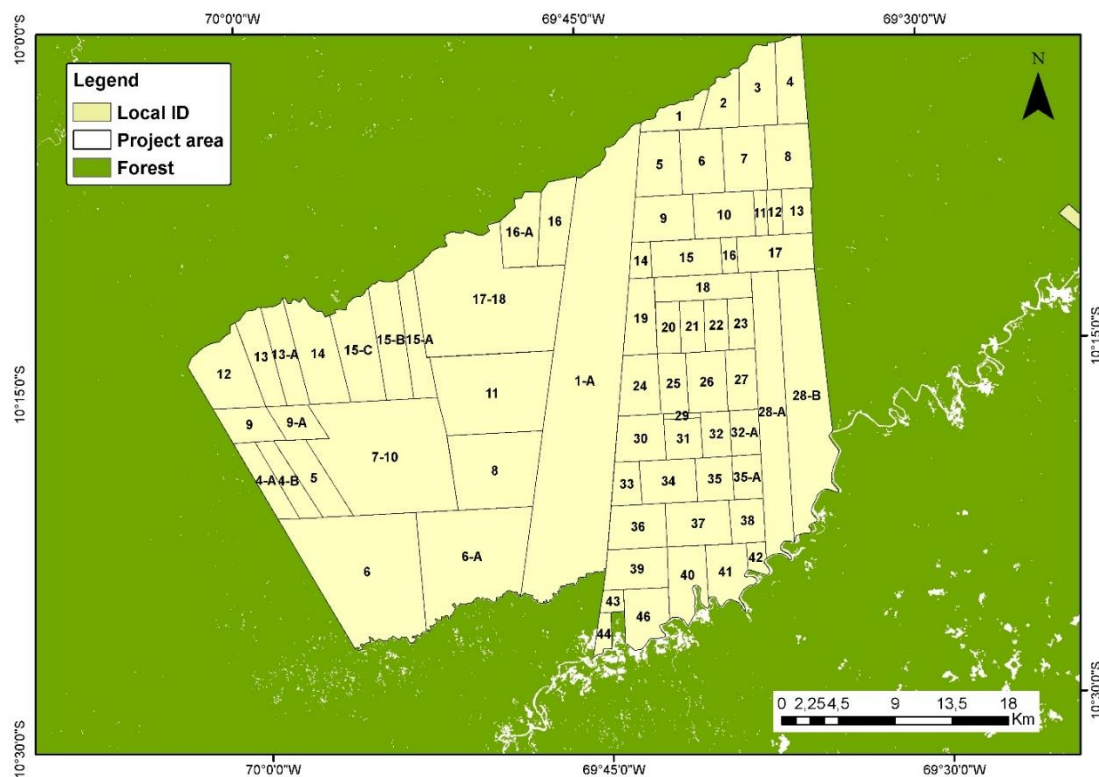


Figure 2 – Indicative Map of the 68 registrations, which are leased by IACO Ambiental Ltda.

1.8 Project Start Date

According to the VCS Standard Version 4.3, the project start date for AFOLU projects is the date on which activities that lead to the generation of GHG emission reductions or removals are implemented.

Therefore, the project start date for both AUD and APD components of the IACO REDD+ Project is defined as 16 December 2021, the date of the legal registration of the Business Partnership Agreement (“Contrato de Parceria Empresarial”, in Portuguese), which has the documented objective of implementing the REDD+ Project at Fazenda Guanabara/Petrópolis. Also, on 16 December 2021 a “Rural Lease Agreement for Environmental Management and Reforestation Purposes” (“Contrato De Arrendamento Rural Para Fins De Gestão Ambiental E Reflorestamento”, in Portuguese) was signed between the landowners and the partners of IACO Ambiental for the purpose of environmental management, forest maintenance, reforestation and commercial exploitation of the respective carbon credits.

1.9 Project Crediting Period

The Project Crediting Period will be 30 years, renewable for another 30 years. The first crediting period will be from 16 December 2021 until 15 December 2051, both days included. The maintenance of project activities throughout the proposed crediting period is ensured by a lease agreement of all parcels of the Fazenda Guanabara/Petrópolis located in the IACO REDD+ Project area to IACO Ambiental Ltda. This contract is defined as "Rural Lease Agreement for Environmental Management and Reforestation Purposes" (or in Portuguese “Contrato de Arrendamento Rural para Fins de Gestão Ambiental e Reflorestamento”) and will be in force for a period of 60 (sixty) years, starting on December 16, 2021, with final term on December 15, 2081.

1stbaseline period: 16/12/2021 to 15/12/2027;

2ndbaseline period: 16/12/2027 to 15/12/2033;

3rdbaseline period: 16/12/2033 to 15/12/2039.

4thbaseline period: 16/12/2039 to 15/12/2045;

5thbaseline period: 16/12/2045 to 15/12/2051;

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	
Large project	X

Estimates of GHG emissions reductions were made on a preliminary basis, assuming a deforestation rate of 0.931% for avoided unplanned deforestation activity, estimated by the deforestation rate of reference region (RRD) in the period from 2012 to 2020. For avoided planned deforestation activity, the GHG emissions reduction estimates take place on the first 7-year deforestation cycle and covered 4,649.33 hectares per year in the first 6 years, and 1,812.20 hectares in the last year.

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
2022	1,769,604
2023	1,784,972
2024	1,800,336
2025	1,815,696
2026	1,831,051
2027	1,846,402
2028	1,028,137
2029	501,930
2030	502,379
2031	502,824
2032	484,140
2033	465,491
2034	446,876
2035	428,297
2036	409,787
2037	391,276
2038	381,884
2039	378,329
2040	374,807
2041	371,318
2042	367,861

2043	364,437
2044	361,044
2045	357,715
2046	354,351
2047	351,052
2048	347,784
2049	344,546
2050	341,339
2051	338,161
Total estimated ERs	21,043,825
Total number of crediting years	30
Average annual ERs	701,460

1.1.1 Description of the Project Activity

The main purpose of the IACO REDD+ project is the conservation of the Amazon rainforest area in a property situated in the municipality of Sena Madureira from planned and unplanned deforestation, center of the Brazilian state of Acre. This will be possible through income generation of carbon credits, and its posterior commercialization, for project proponents.

This project will be implemented to curb planned and unplanned deforestation on private property. Since 2013, the landowners of the Fazenda Guanabara/Petrópolis parcels have plans for planned deforestation as permitted by national law 12,651 (more details on section 3.4). In 2015, a request for forest cover suppression was filed with IMAC: Instituto do Meio Ambiente do Acre (Acre Environmental Institute, in English) to obtain permission to suppress native vegetation in 20% of the property's area. The purpose of this suppress was the timber exploration and the creation of pastures, a much-needed income and commonly seen activity at the region where the project is located. Between 2015 and 2019, the project proponents consulted different service providers in order to ask for proposals for infrastructure and labor for the implementation of the aforementioned activities. In 2019, the landowners received a proposal for a commercial partnership with a logging company. Due to the beginning of the pandemic, it was only in 2021 that the landowners continued the planned deforestation plan with the request of a proposal for an environmental service of suppression for the purpose of logging and analysis of pasture production. In the absence of the project, 29,708,23 hectares, or 20% of Fazenda Guanabara/Petrópolis would have been deforested.

The other 80% of the property would be under high pressure from unplanned deforestation due to the insufficient presence of activities at the project site and the lack of a surveillance budget, which facilitates encroachment by subsistence farmers and ranchers. As shown in *Figure 3* part of the property was illegally deforested by these same agents in the last 5 years.

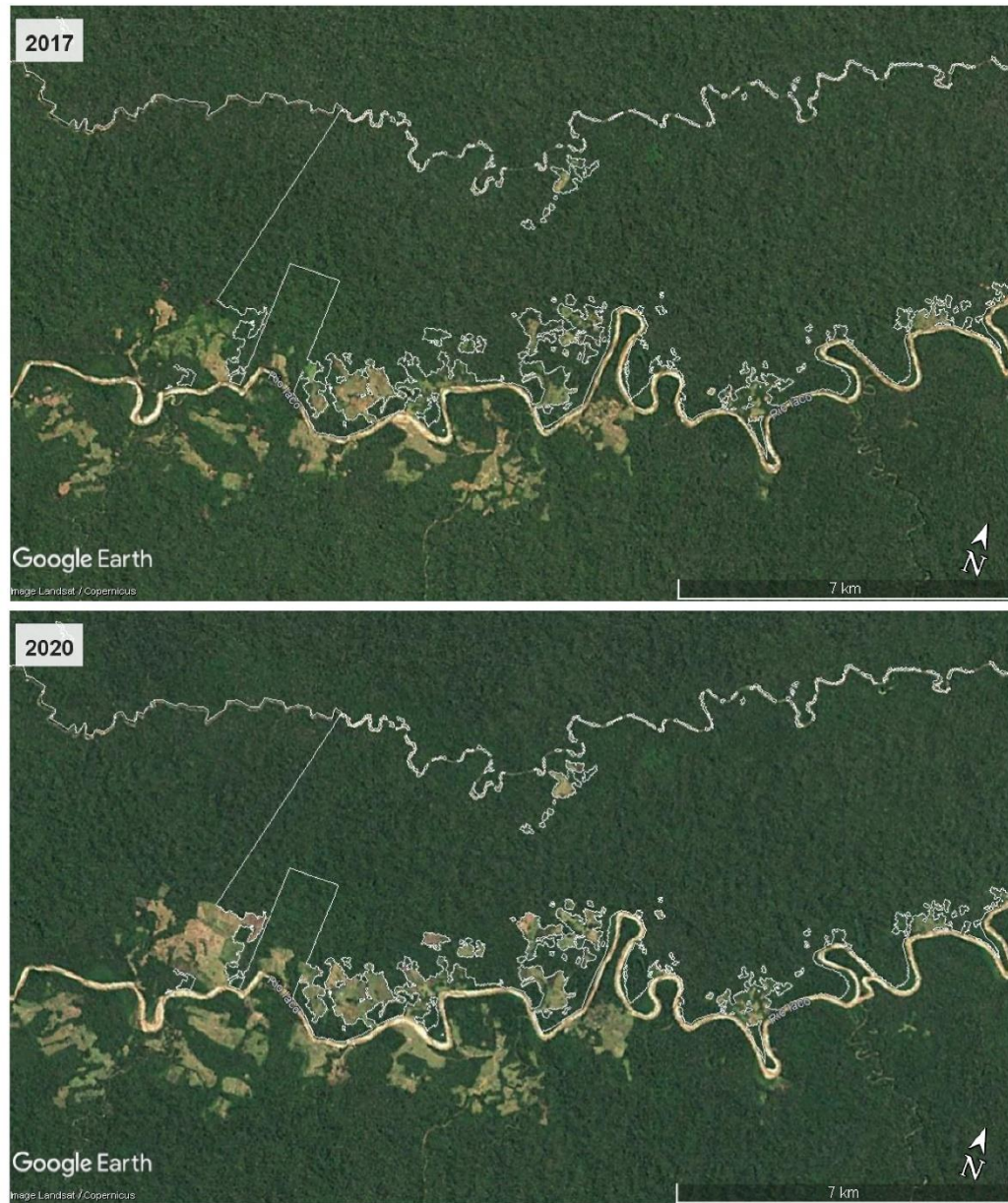


Figure 3 - Deforestation events occurred in 2017 and 2020 in the project area surroundings.

To mitigate these factors and achieve the reduction of emissions from deforestation, the IACO REDD+ Project has planned the following strategies in different spheres:

Climate sphere:

- a) Installation of signboards to inform the presence of the REDD+ project. As the project area boundary is not sufficiently recognized by community and immigrants, these signboards are intended to protect the project area against invasion and illegal deforestation and will contain information about the project and contact with the project proponents. These signboards will be allocated along the banks of the river since the main vector of deforestation comes from the Iaco River.
- b) Forest patrols. These patrols will have the objective of enforcing environmental laws, preventing the action of illegal agents of deforestation and/or degradation, as well as the maintenance of vegetation cover and conservation of biodiversity. It can also improve the relationship of trust between project proponents and local communities by involving residents in this activity. Such patrols will take place in the project area and in the leakage belt. New 4x4 car will be purchased (Figure 4), to ensure security at the project and surrounding areas; During the dry season, the risk of fire spreading in the project area and its leakage belt increases due to fires to open new areas. Therefore, the patrol frequency and team would be intensified as necessary at these times. If any evidence of encroachment, illegal logging or exploitation or fire outbreaks is found, the situation should be shared with the patrol team personnel, and immediately reported to the project management team for action to be taken and duly informed as per the definition of loss events.



Figure 4 - Car to be used by the patrol crew.

- c) Basecamp construction. This basecamp will be open to residents surrounding the project area, providing a safe place for consultation and to exchange opinions. In addition, this basecamp will have facilities such as a kitchen, bathroom and bedroom that can be used by visitors and project employees. Safety and communication equipment used by patrols such as leggings, radio and GPS, as well as fire extinguishing equipment (bombs, extension hoses, fire boots, etc.) will also be available there.
- d) Fire Prevention and Extinguishing. In the IACO REDD+ project region, forest fires are associated with human activity and occur with greater frequency, intensity, and severity in the driest periods and in areas with certain levels of occupation and/or degradation. Fire brigade training will be given to all people who show interest and vocation and who live close

to the project area; Instruments to combat fires will also be made available in the project basecamp. Training related to awareness of fire ecology and good practices in preparing areas for cultivation, with a focus on preventing the spread of fire in forests will also be a project activity and will be open to anyone who wants to participate.

- e) High resolution satellite imagery for monitoring forest cover and biodiversity conservation. There are no roads within the project area and access is limited to boats via rivers and waterways (*Figure 5*). Therefore, the area that can be monitored by forest patrol is limited to the banks of the Iaco River, using satellite images for monitoring and observation of other areas.

Social sphere:

- f) Regulation of land use in surrounding project area. The objective is to provide the demarcation of areas destined for the population. Currently the best ways to carry it out are being studied and will be informed at the complete PD for registration and VVB assessment.
- g) Training courses. Training will be offered on different topics, but especially on environmental awareness and climate change, biodiversity and ecosystem services and sustainable management of forests, agricultural lands, and grasslands. These trainings will be open to anyone who is interested and will be disseminated to the entire local community and neighbors to the project area.
- h) Employment: People residing around the IACO REDD+ project area will be contracted for some of the project activities, such as forest patrol, base construction, forestry guides and communication. These contracts will be made on a daily paid basis for the elaboration of a specific service. According to local interviews, this is the preferred way for most people residing in the area.
- i) Education and health: Conversations and interviews with stakeholders are ongoing to define the activities that best serve the community. So far, the IACO REDD+ project will support the improvement of education in the municipality of Assis Brasil. It is important to clarify that although the project is located at the municipality of Sena Madureira, Assis Brasil has the closest settlements to the project. Regarding health, the establishment of telemedicine at the health post in the municipality of Assis Brasil is already being discussed. Together with telemedicine, the IACO REDD+ project will provide regular training to nurses and improve access to the local internet so that telemedicine consultations works well and meet community needs. Furthermore, the project aims to contact at and make contracts with doctors from the cities nearby so it will be possible to the communities to have face to face consultations and treatments.
- j) Improvement and diversification of the local economy. The IACO REDD+ project will provide technical assistance to support existing agricultural activities and promoting local culture by encouraging handicrafts for women residents in project region. Project proponents are committed to complete transparency and will not benefit financially from sales.
- k) Transparency: The project proponents are committed to share information regarding project objectives and results achieved through booklets and visits by project employees.
- l) Infrastructure. The project will provide the installation of structures of sewage treatment in the public school and the health post that serve the population.

Biodiversity sphere:

- m) Monitoring of flora and fauna. The photographic record, for instance, can document, record, verbalize and trigger nature conservation, wildlife protection and environmental action. By hiring a photographer with extensive experience in recording fauna and flora, the IACO REDD+ project intends, in addition to registering local biodiversity, to be closer to threatened species, capture changes in the environment and promote nature conservation.
- n) Partnerships with universities and research institutions. the project area will be offered to be used to the study of forest dynamics, biodiversity, conservation, and other themes related to the area. The project will provide the infrastructure necessary so that researchers can stay in the area such as forest guides to accompany them.

Jurisdictional and Nested REDD (JNR)

The REDD+ IACO project is located in the state of Acre, which has a State System of Incentives for Environmental Services focused on promoting the reduction of GHG emissions, in accordance with the voluntary target of the State of Acre, the SISA-Carbon of 2010. However, there are no obligations for private REDD projects in adhesion to this jurisdictional program of the State of Acre and there is no risk of double-counting since the Government of Acre does not have a jurisdictional program validated by a carbon market standard. To date, the Acre SISA-Carbon is a public administration document that serves as a guide, which allows others to voluntarily join the Program if they choose. More details on compliance of IACO REDD+ Project and SISA-Carbon area available in section 1.14.

1.12 Project Location

The IACO REDD+ Project is situated in Sena Madureira, center of the Brazilian state of Acre. The municipality's area is 25,278 km², representing 16.5% of the State area. The project area is located between the Iaco and Macauã rivers within a private property, named "Fazenda Guanabara/Petrópolis", which has 148,541.15 hectares.

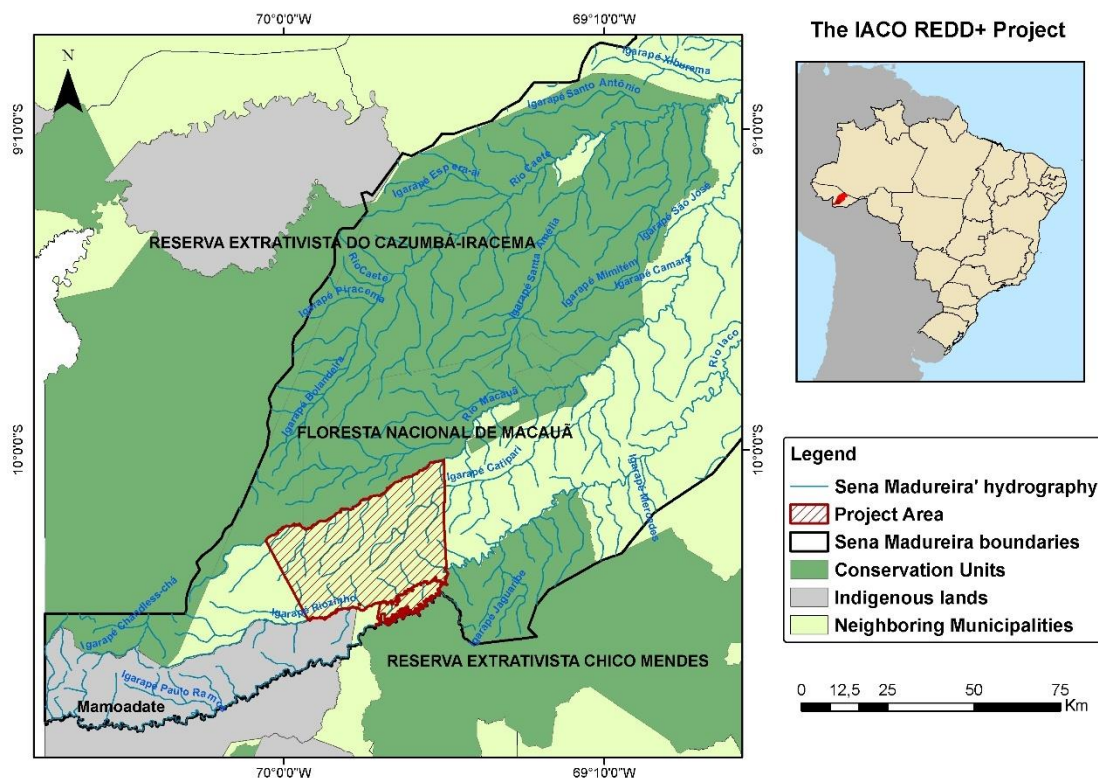


Figure 5 – IACO REDD+ Project location. The conservation units and indigenous lands that surround the area are highlighted. Source: FUNAI⁶, IBGE⁷ (2022) and ANA⁸ (2018).

The access to the project area is difficult depending on the season and weather conditions. By watercourses, up the Iaco River, the project area is approximately 30 hours from center of Sena Madureira city. By land, it is possible to reach the project area from Rio Branco, capital of Acre State, by the State Highway AC – 90, following the Nova Olinda and Guanabara branches (it is not georeferenced in OpenStreetMap⁹ and Google Maps¹⁰). From Assis Brasil city, the city closest to the project's settlements, it is possible to reach the project area after 77 km of road, approximately three and a half hours. For the road trip, a 4x4 car is needed.

The project area is surrounded by conservation units. On the north, the project area borders the Flona Macaúã (approximately 176,349.02 hectares) and the Reserva Extrativista Cazumbá-Iracema (approximately 754,987.10 hectares). On the south, the Mamoadate Indigenous Lands

⁶ Fundação Nacional do Índio (FUNAI), <https://www.gov.br/funai/pt-br>

⁷ Instituto Chico Mendes da Conservação da Biodiversidade (ICMBio), <https://www.gov.br/icmbio/pt-br/servicos/geoprocessamento/mapa-tematico-e-dados-geoestatisticos-das-unidades-de-conservacao-federais>

⁸ Agência Nacional de Águas, <https://metadados.snirh.gov.br/geonetwork/srv/api/records/0f57c8a0-6a0f-4283-8ce3-114ba904b9fe>

⁹ OpenStreetMap, <https://www.openstreetmap.org/>

¹⁰ Google Maps, <https://www.google.com.br/maps>

(approximately 312,584.49 hectares) and the Reserva Extrativista Chico Mendes (approximately 916,917.93) are neighboring the project.

A KML file is attached separately to better indicate the coordinates of the project.

1.13 Conditions Prior to Project Initiation

General characteristics of the project area and reference region

The land within the Project Area has original forest cover from the amazon rainforest, which has been intact for over 10 years (*Figure 1*). However, the project region is experiencing deforestation pressure increase driven primarily by extensive, small scale beef cattle farming and subsistence colonization farms (see *Figure 3*). In the IACO REDD+ Project region the deforestation for these purposes occurs mainly along major roads and along riverbanks.

Geology and Geomorphology

According to ZEE (2011)¹¹, the Acre state is located in nine geomorphological units (*figure 6*): a Planície Amazônica, Depressão do Endimari-Abunã, Depressão do Iaco-Acre, Depressão de Rio Branco, a Depressão do Juruá-Iaco, Depressão do Tarauacá-Itaquai, Depressão Marginal à Serra do Divisor, Superfície Tabular de Cruzeiro do Sul e Planaltos Residuais da Serra do Divisor.

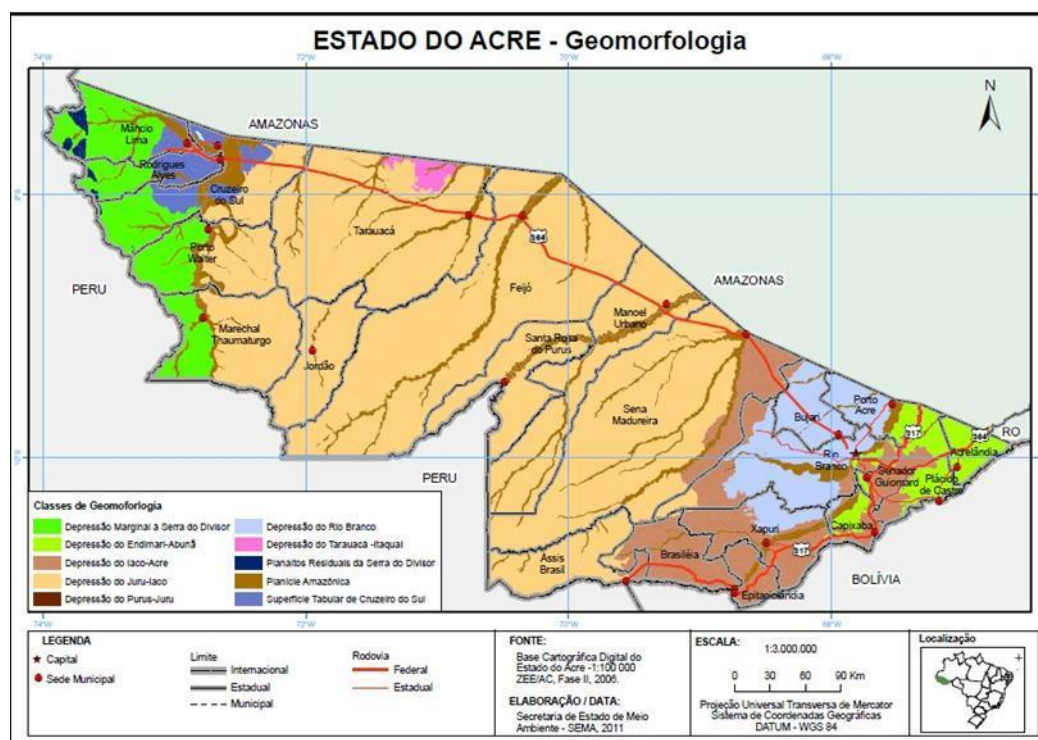


Figure 6 – Geomorphological units found in the Acre state. Source: ZEE (2011)

The main geotectonic unit is the Acre Basin, delimited by the Arc of Iquitos and the Andean belt. The basin is predominantly formed by poorly consolidated sedimentary material of the Cenozoic

¹¹ Zoneamento Ecológico-Econômico fase II. Available at: < <http://semapi.acre.gov.br/zee-acre/>>.

age (Resende & Machado, 1988)¹². The local landscape presents marked geology influences, tectonic activity, and physicochemical activities in its formation and evolution (Kronberg et al., 1989)¹³. The area is constituted by shallow valleys, and the soil is predominantly sedimentary with high sediment cycling. Soils presents elevated contents of high activity clay and silt. It is therefore a pedologically recent region (Kronberg & Benchimol, 1992)¹⁴.

Topography and Soil

We obtained topography from SRTM mission Shuttle Radar Telematic Mapper data (INPE's TOPODATA Program ¹⁵) resampling in a resolution of 30m. Altitude data were used in relation to sea level and slope in %.

In general, the Acre state presents a gentle topography¹⁶ with a slope of less than 15%. The altimetric dimensions range from 118 to 621 meters from sea level. The average altitude in the project area is 255 m above sea level, ranging from 203 to 309 m.

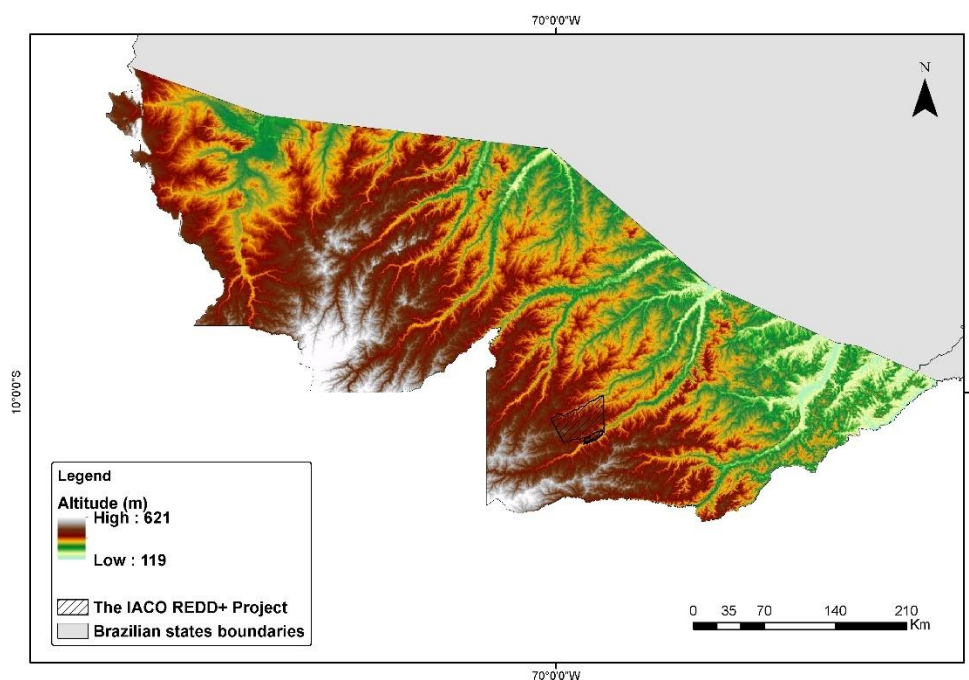


Figure 7 – Altitude in the Acre state. Source: INPE.

¹² Resende & Machado, 1988. Cotas fluviométricas do rio Acre, suas causas e implicações na política de colonização. Available at: < <https://www.scielo.br/j/aa/a/BWskdvkz4gNnQTCv5mdb3D/?lang=pt> >

¹³ Kronberg et al, 1989. Geochemical variations in Solimões formation sediments (Acre basin, Western Amazonia). Available at: < <https://www.scielo.br/j/aa/a/r9S8wMtPMmSq3Rxd3xfKM/abstract/?lang=en> > Available at: < >

¹⁴ Kronberg & Benchimol, 1992. Geochemistry and geochronology of surficial acre basin sediments (western amazonia): key information for climate reconstruction. Available at: < <https://www.scielo.br/j/aa/a/qRbKRkgQ5zsWXpLWWYDZLSv/?lang=en> >

¹⁵ National Institute of Spatial Research (INPE). Available at <http://www.dsr.inpe.br/topodata/>

¹⁶ Bardales, 2009. Estratificação ambiental, classificação, mineralogia e uso do solo da microbacia do Igarapé Xiburema, Sena Madureira, Acre. Available at: < <https://www.locus.ufv.br/handle/123456789/1598> >

The National Spatial Data Infrastructure (INDE) provides a Brazilian environmental database. This infrastructure has a variety of information obtained by government agencies; this information can be disposed of spatially by clicking on the desired area. From IBGE data, at a scale of 1:250.000¹⁷, we found that the project area soils are predominantly composed of Haplic Luvisol (63,85%), followed by Red Yellow Argisol (30,57%), Haplic Gleysol (5,57%) and Chromic Luvisol (0,001%).

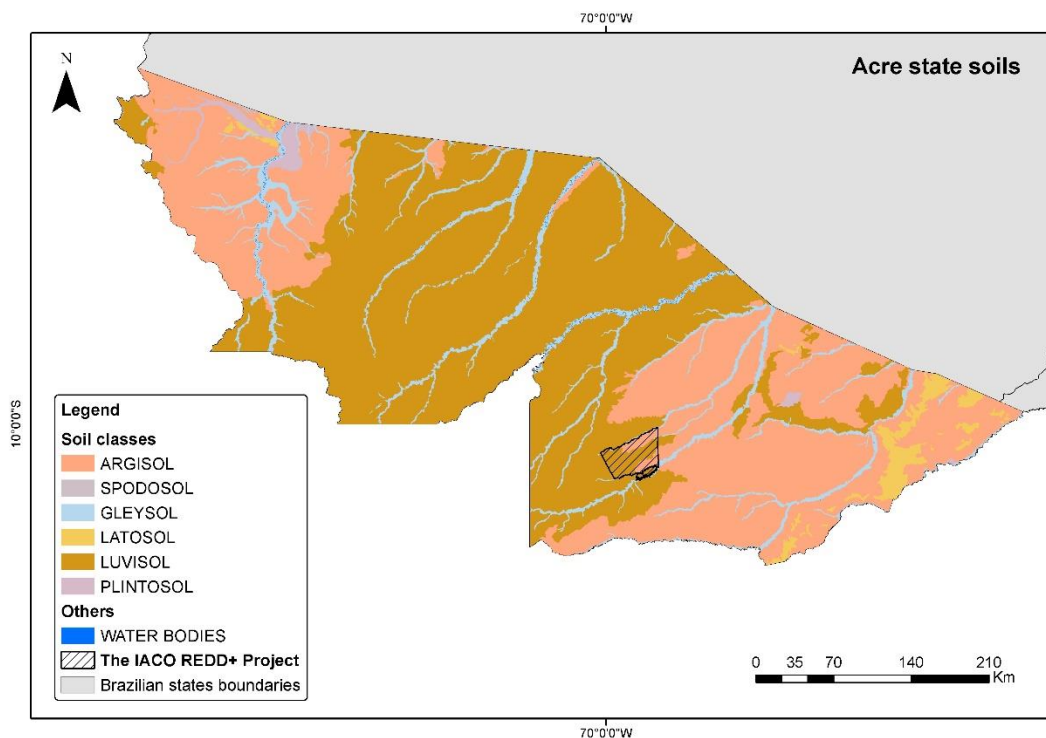


Figure 8 – Soils found in Acre state. Source: IBGE (2021)¹⁸

As described above, the soils that occur within the project boundary are not inserted in the category of organic soils, being eligible for project activity in accordance with the applied methodologies in this PD.

Climate and Hydrography

The typical climate in the state of Acre is the humid equatorial climate (Af category in Köppen Climate Scale)¹⁹ which is predominantly low thermal amplitude with high temperatures, average temperature around 25°C; and high precipitation levels, usually around 2000mm (Bardales, 2005)²⁰. Usually, the rainy season extends from November to April and the dry season from June

¹⁷ Brazilian Agricultural Research Corporation (EMBRAPA), <http://www.cnps.embrapa.br/>

¹⁸ Brazilian Institute of Geography and Statistics (IBGE), <https://www.ibge.gov.br/geociencias/>

¹⁹ Alvares, C.A., Stape, J.L., Sentelhas, P.C., Gonçalves, J.L.M.; Sparovek, G. Köppen's climate classification map for Brazil. Meteorologische Zeitschrift, v. 22, n. 6, p. 711-728, 2013.

²⁰ Bardales, 2005. Gênese, morfologia e classificação de solos do baixo vale do Rio Iaco, Acre, Brasil. Available at: <https://www.locus.ufv.br/bitstream/123456789/10964/1/texto%20completo.pdf>.

to September. Rivers and streams show high seasonality occasioned by the low relatively depth of regional soil, which occurs due to presence of high activity clays in the soil. Although there are high levels of precipitation, the water has difficulty to infiltrate in the soil (Resende & Pereira, 1988)²¹.

The project is located at the west bank of Iaco River, an affluent of the Purus River, which in turn is an affluent of the Solimões River (figure 5). The Iaco River has an extension of approximately 650 km and two main affluent rivers, Macauã and Caeté Rivers (Pereira & Morais, 2015)²².

Socio-economic conditions

The project is in Sena Madureira municipality, in the state of Acre. Sena Madureira has an estimated population of 47,168 people (2020), of which 5.1% were occupied under a registered employment regimen, for which the average monthly salary was of 1.5 minimum salaries. Almost 47% of the residences counted with only half of a minimum salary (IBGE, 2020).

The municipal GDP per capita of Sena Madureira is R\$ 11.783,19 (figure 9), ranked in 3861^a position in Brazil (IBGE, 2020). Almost 17% of GDP come from agriculture and livestock activities.

In Sena Madureira, the agricultural production involves the cultivation of temporary and perennial crops such as avocado, papaya, coffee, coconut, lemon, orange, passion fruit, pineapple, sugarcane, cassava, watermelon and others (see more details in <https://cidades.ibge.gov.br/brasil/ac/sena-madureira/pesquisa/14/10193>).

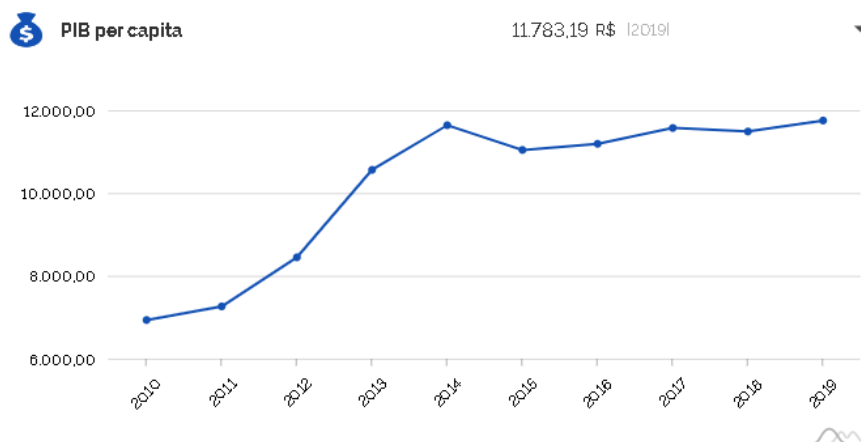


Figure 9 – GDP per capita of Sena Madureira from 2010 to 2019. Source: IBGE (2020).

Figure 10 shows how livestock and the number of cattle heads have been increasing in Sena Madureira, since 2004. It is possible to observe the ascendant curve in the graphic rising by

²¹ Resende & Pereira, 1998. Cotas fluviométricas do rio Acre, suas causas e implicações na política de colonização. Available at :< <https://www.scielo.br/j/aa/a/BWskdvkz4gNnQTCv5mdb3D/?lang=pt>>.

²² Pereira & Morais, 2015. Técnicas de geoprocessamento aplicadas aos problemas ambientais que afetam o rio Iaco dentro do limite do município de Sena Madureira-AC. Available at:< <https://periodicos.ufsm.br/reget/article/view/13867/pdf> >.

161% in the last 10 years. In 2020, there were around 364.5 thousand heads (IBGE, 2020). This number meets will the second highest deforestation rate found in Acre in 2021, where Sena Madureira and Feijó together were responsible for 40% of deforestation in the state (Imazon, 2022)²³.

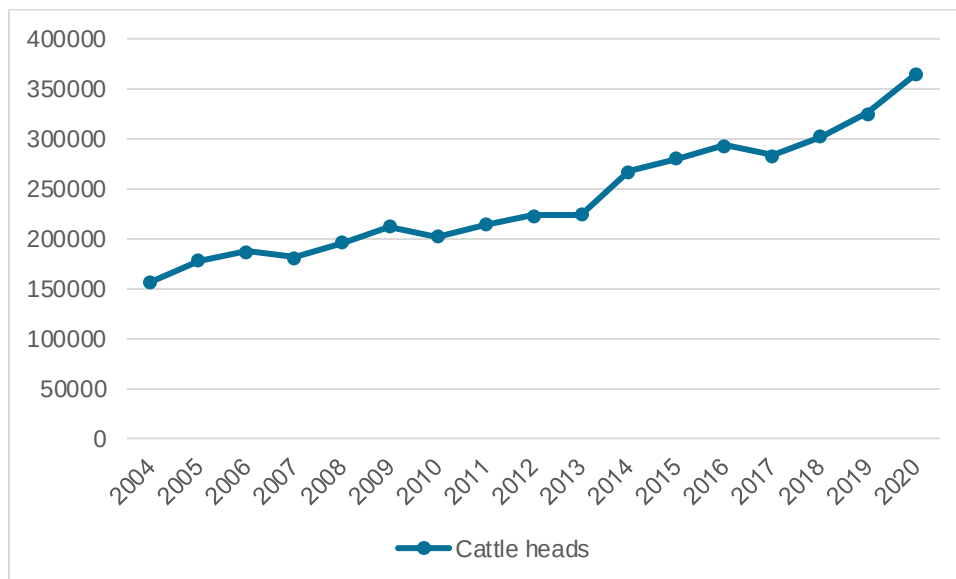


Figure 10 – Number of cattle heads in Sena Madureira from 2004 to 2020. Source: IBGE (2021).

The Human Development Index (HDI) of Sena Madureira is 0,603. It means that the average achievement in key dimensions of human development is medium, considering the index aspects such as long and healthy life, being knowledgeable and have a decent standard of living. In this sense, we found poor information about health and sanitation; education and, security for Sena Madureira, for example.

In terms of health and sanitation, there is not any published public data for Sena Madureira. Thus, we found that only 55,2% state domiciles in Acre had access to the supply network of treated water, 40,6% had sewage treatment, and just 68,7% had any form of residue collection (ZEE, 2021)²⁴.

According to the IBGE (2021), in 2019, Sena Madureira was evaluated with rates of 5,7 for public elementary school and an average of 4,8 for high school by the Brazilian Education Ministry. These values refer to Basic Educational Development Index (Ideb) that evaluates each municipality and state according to the students' permanence in school, obtained from the School Census, and from the performance averages in the Basic Education Assessment System (Saeb). This index generates numerical evaluations between 0 and 10, with 10 being the best possible scenario. As for analphabetism rates, although the Acre state is reducing the population's inability to read and write, their number is still high, and the official information and

²³ Imazon. Available at: <https://imazon.org.br/imprensa/desmatamento-na-amazonia-chega-a-1-606-km2-em-agosto-maior-area-da-decada-no-mes/>

²⁴ Acre, 2021. Ecological and economic zoning, Acre. Available at: < <https://ipam.org.br/bibliotecas/fase-iii-do-zoneamento-ecologico-economico-do-estado-do-acre/>>

evaluating index are outdated. Still, in 2012 almost 14% of the population with more than 15 years old were illiterate, and according to ZEE, this data remained alarming in 2021.

Vegetation and Biodiversity

The IACO REDD+ project is covered by Lowland Open Forest (94,52%) and Alluvial Open Forest (5,48%)²⁵. The Alluvial Forest usually follows the margin of the watercourses and does not present topography variation. Yet the Lowland formation usually occurs in coastal plains at altitudes less than 100m. All these vegetation types belong to a very sensitive area due to fauna and flora conservation targets found there²⁶ (figure 5).

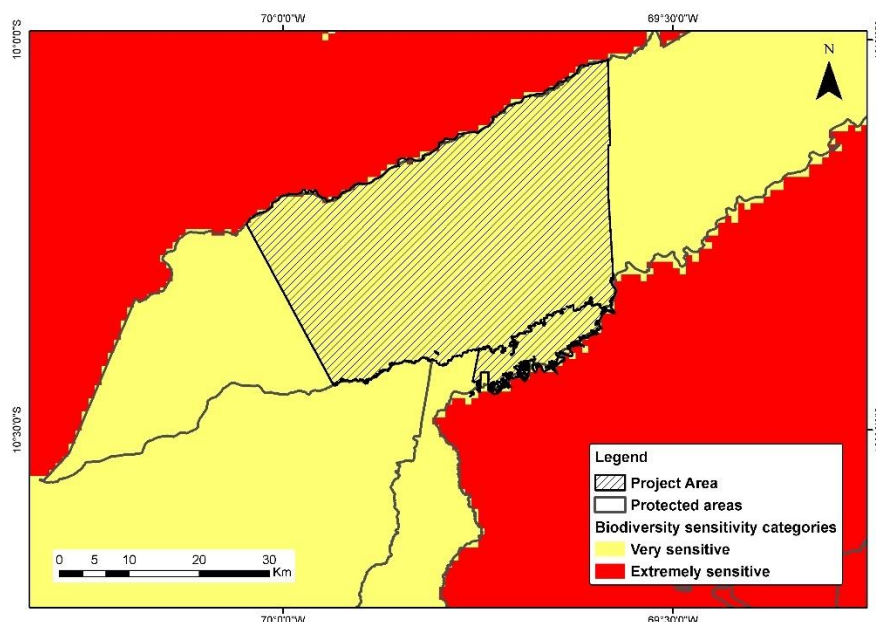


Figure 11 – The IACO REDD+ Project is located in zones classified as Very Sensitive, according to Plan for Reducing the Impact of Land Road Infrastructures on Biodiversity developed by ICMBio.

According to PRIM (Plan for Reducing the Impact of Land Road Infrastructures on Biodiversity) developed by ICMBio (Instituto Chico Mendes de Conservação da Biodiversidade) in 2018, conservation target species of flora and fauna are extinction-threatened species (table 2)) that go beyond impacts of land road infrastructure, and they could be considered good examples for prevention, mitigation and/or protection actions against negative impacts on ecosystems in general.

²⁵ Brazilian Institute of Geography and Statistics (IBGE), 2021. Sena Madureira. Available at: <<https://bdiweb.ibge.gov.br/#/home>>. Last access: 24/06/2022.

²⁶ ICMBio (Instituto Chico Mendes de Conservação da Biodiversidade), 2018. Plan for Reducing the Impact of Land Road Infrastructures on Biodiversity (PRIM). Available at: <<https://www.gov.br/icmbio/pt-br/assuntos/biodiversidade/planos-de-reducao-de-impacto/prim-ivt>>. Last access: 24/06/2022.

Table 2 – Conservation target species, according to Plan for Reducing the Impact of Land Road Infrastructures on Biodiversity (PRIM – ICMBio)

Biodiversity Component	Group or Family	Species
Fauna	Birds	<i>Alectrurus tricolor</i>
		<i>Asio stygius</i>
		<i>Coryphaspiza melanotis</i>
		<i>Cranioleuca vulpecula</i>
		<i>Crax fasciolata pinima</i>
		<i>Frederickena unduliger</i>
		<i>Hydropsalis anomala</i>
		<i>Lophotriccus eulophotes</i>
		<i>Phaethornis aethopygus</i>
		<i>Sporophila maximiliani</i>
		<i>Strix virgata</i>
		<i>Syndactyla ucayalae</i>
		<i>Urubitinga coronata</i> ;
	Mammals	<i>Ateles chamek</i>
		<i>Atelocynus microtis</i>
		<i>Blastocerus dichotomus</i>
		<i>Chrysocyon brachyurus</i>
		<i>Lagothrix cana cana</i>
		<i>Leopardus wiedii</i>
		<i>Lycalopex vetulus</i>
		<i>Myrmecophaga tridactyla</i>
		<i>Panthera onca</i>
		<i>Priodontes maximus</i>
		<i>Puma concolor</i>
		<i>Puma yagouaroundi</i>
		<i>Tapirus terrestris</i>
Flora	Fabaceae	<i>Apuleia leiocarpa</i>
	Lecythidaceae	<i>Bertholletia excelsa</i>
	Lauraceae	<i>Mezilaurus itauba</i>
	Meliaceae	<i>Swietenia macrophylla</i>
	Myristicaceae	<i>Virola surinamensis</i>

In addition, there is a zone with high conservation priority of 6.209,89 ha (**Figure 12**) located inside the project area, according to ZEE (3rd Edition). It means that conservation actions (flora and fauna), sustainable use and benefit-sharing of the Brazilian biodiversity have high priority in these areas.

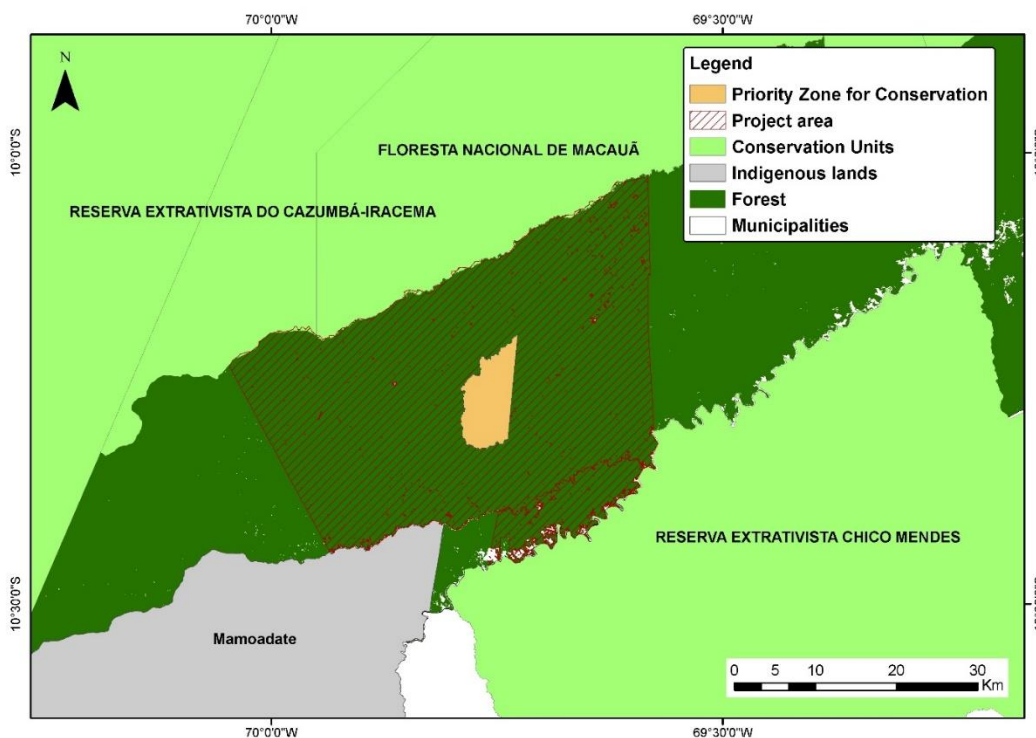


Figure 12 – Zone with high conservation priority within the IACO REDD+. Fonte: ZEE-AC (2021)²⁷.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The IACO REDD+ Project and its proponents is in full compliance with all applicable and relevant national, state, and local laws, including statutes and regulatory frameworks.

National Laws and Regulatory Frameworks

Brazilian Federal Constitution²⁸: In the Chapter 6, Article 225, the Paragraph 1 discusses the ecologically well-balanced environment through the preservation and restoration of essential ecological processes and provide for the ecological management of species and ecosystems, as well as the preservation of the diversity and integrity of the country's genetic heritage and inspect entities dedicated to research and manipulation of genetic material. Further, in Paragraph 4 is stated: “The Brazilian Amazonian Forest, the Atlantic Forest, the Serra do Mar, the Pantanal Mato-Grossense and the coastal zone are part of the national patrimony, and they shall be used, as provided by law, under conditions which ensure the preservation of the environment.” The IACO

²⁷ Informations obtained at the Ecological and Economic Zoning of the state of Acre. Available at: <
http://semapi.acre.gov.br/wp-content/uploads/sites/20/2021/12/COMPLETO-Resumo-Executivo-do-ZEE-Acre-Fase-III_V16_WEB.pdf>

²⁸ Georgetown University, “1988 Constitution, with 1996 reforms in English” Available at:
 <<https://pdba.georgetown.edu/Constitutions/Brazil/english96.html#mozTocId920049>> Last access: 30/04/2022

REDD+ project activities will protect approximately 147 thousand hectares of Brazilian Amazonian Forest, ensuring an ecologically balanced environment and preserving one of the most biodiverse areas in the world ²⁹.

Brazilian Forest Code (Law N° 12.651, 25/05/2012)³⁰: In Chapter 1, Article 1, is stated: “This Law establishes general standards on the protection of vegetation, permanent preservation areas, and legal reserve areas”. It establishes that every rural property located in forested areas should have a Permanent Preservation Area and Legal Reserve areas (LR) of eighty percent (80%) of a rural property in the Brazilian Legal Amazon. In addition, in Chapter 5, Article 26 is stated: The removal of native vegetation to alternative land use, are conditioned to the registration of the property in the Rural Environmental Registry - CAR, mentioned in art. 29, and the prior authorization of the competent state agency Sisnama (Environment National System), corresponding to national and state environmental agencies”. In the case of Acre State, the official agency is the Environmental State Secretariat (SEMA). All properties belonging to the IACO REDD+ project have officially allocated 80% of their total area as LR and all the Permanent Preservation Areas (APPs) are respected. Moreover, all of them are duly registered in CAR. Finally, as all properties included in the IACO REDD+ project are in full compliance with this law, it is expected that any forest conversion must be formally authorized by the competent body.

Brazilian National Policy on Climate Change (Law N° 12.187, 29/12/2009)³¹: The PNMC defines actions and measures aimed at mitigating and adapting to climate change, with the following specific objectives for deforestation: seek a sustained reduction in deforestation rates in all Brazilian biomes, until reaching zero illegal deforestation and eliminating the net loss of area of forest cover in Brazil by 2020. The IACO REDD+ Carbon Project's primary purpose is to avoid the GHG emissions from the deforestation of 147,398 hectares of original Amazon rainforest with estimated GHG emission reductions over the 30 years of project lifetime, are of 21,043,825 tCO₂e (annual average of 701,460 tCO₂e), contributing with climate change mitigation.

State Laws and Regulatory Frameworks

Decree N°. 7.734 of 06/06/2014³²: This Decree, consisting of 4 Chapters divided into 12 articles, regulates the Rural Environmental Registry (CAR) within the State of Acre. In the Chapter 3, Article 4, is stated: “All rural properties existing in the State of Acre, including those that already have the Rural Environmental Licensing - LAR and those that extend to other States, must be registered in the CAR”. All properties (registrations) incorporated into the IACO REDD+ project are in full compliance with the CAR in the Sicar-Acre system.

SISA Acre (Law n°. 2.308 /2010) and Jurisdictional Program of the State of Acre (2013): “The Acre Environmental Services Incentive System – SISA is linked to the Forest Environmental Asset

²⁹ Flores, M. et al. WWF's Living Amazon Initiative (Grambs Corporación Gráfica, 2010).

³⁰ BRASIL. Law nº. 12.651, of 25 May 2012. Forest Code. Diário Oficial [da] República Federativa do Brasil, Brasília, DF, 25 May 2012. Available in English at: < https://www.gov.br/mj/pt-br/acao-a-informacao/atuacao-internacional/legislacao-traduzida/lei-no-12-651-de-25-de-maio-de-2012-senasp_eng-docx.pdf>. Last access: 30/04/2022

³¹ Law No. 12.187 instituting the National Policy on Climate Change (PNMC). Available at < <https://www.braziliannr.com/brazilian-environmental-legislation/law-no-12187-brazilian-national-policy-on-climate-change>>. Last access: 02/06/2022

³² Decree nº 7734 of 06/06/2014 regulating the Rural Environmental Registry (CAR) within the State of Acre. Available at < <http://extwprlegs1.fao.org/docs/html/bra137574.htm>>. Last access: 03/05/2022

Valuation Policy, which is composed of a set of interconnected programs, projects and actions, whose purpose is the conservation and improvement of environmental quality throughout the state.” In one of SISA Acre programs, the ISA Carbon program, which will be implemented under a fully jurisdictional approach, at the subnational scale, intends to incorporate incentives and projects related to reducing emissions from deforestation and forest degradation – REDD+. In the SISA Acre’s description, particularly in articles 22 and 23, there is no provision of any determination, rule, limitation, or specification on the execution of REDD projects in the Acre State territory. In the articles 24, 25 and 26 there is a mention on the establishment of a preliminary period and commitment to the emission reduction target, as well as the establishment of a limit of registrable carbon units for special projects, and the intention to implement a system of accounting for reductions. However, there is no definition or regulation on the application of these themes. Thus, the SISA-Carbon Program does not have legal applicability to REDD projects until all the issues mentioned above are fully regulated. In addition, there is no prohibition on carrying out these projects by the private sector, nor the obligation for private REDD projects to adhere to the SISA-Carbon Program. Regarding the Acre REDD jurisdictional program, it is a public administration document that serves as a guide, which allows others to voluntarily join the Program if they choose. Moreover, it was not formally validated under the VCS, or any other standard of REDD+ in Brazil or worldwide. Therefore, there are no obligations for the IACO REDD+ Project in adherence to the jurisdictional program of the State of Acre and there is no risk of double-counting since the Government of Acre does not have a jurisdictional program validated by a carbon market standard.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered, nor is it seeking registration under any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The activities that result on the GHG removal of the project are not included in any other emission trading programs or mechanisms that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

The activities that result on the GHG removal of the project are not included in any other emission trading programs or mechanisms that includes GHG allowance trading.

1.17 Sustainable Development Contributions

The IACO REDD+ Project is responsible for more than just mitigating climate change by avoiding carbon emissions from deforestation. The project will also ensure the conservation of a large biodiversity hotspot area and implement a series of social activities for the benefit of the local community.

It is also important to highlight how the project activity is in line with the nationally determined Contribution, conceived under the Paris Agreement, which outlines the efforts each country is

willing to make in order to reduce national emissions and adapt to the impacts of climate change. The Brazilian NDC has a specific theme for the forestry and land use change sectors³³, in which, of the 5 goals established, the IACO REDD+ project directly contributes with two:

- The importance of enforcing the forestry code, which has already been observed and resulted on further efforts on conservation and restoration of legal reserves and permanent protection areas.
- The importance of enforcing the end of illegal deforestation at the Amazonian biome and grant the compensation of emissions caused by legal vegetation suppression until 2030.

The project benefits are also in line with the SDGs (Sustainable Development Goals), which were nationalized³⁴. The IACO REDD+ project activities can contribute to the sustainable development of the following socioeconomic and environmental issues.

SDGs	SDGs description	Project activity
	Ensure healthy lives and promote well-being for all at all ages	The population living nearby the project area already has access to a health post (4 km from the community). The project aims to install internet points at the post in order to enable online consultations and healthy preventive education. In addition, the project aims to sign agreements with doctors from neighboring cities so that face-to-face care can be provided, especially for women going through motherhood.
	Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all	Construction of support points for studies in the preserved area, structures that allow researchers to stay (bathroom, bedroom, kitchen, study room). The project aims to contact universities as a way of encouraging research in the area.

³³ MMA. REDD+ e a NDC do Brasil, 2019. Available at: < <http://redd.mma.gov.br/pt/redd-e-a-indc-brasileira>>. Last Access: 20th October. 2021.

³⁴ IPEA. ODS – Metas Nacionais dos Objetivos de Desenvolvimento Sustentável. 2018. Available at: < https://www.ipea.gov.br/porta/images/stories/PDFs/livros/livros/180801_ods_metas_nac_dos_obj_de_desenv_susten_prop_os_de_adequa.pdf>. Last access: 20th October 2021.

	Ensure availability and sustainable management of water and sanitation for all	The project aims to install sewage treatment structures adapted to rural conditions in public places (regional school and health center), this installation will be accompanied by the training of operators of these systems. Some alternatives are septic tanks followed by anaerobic filters, evapotranspiration tank or even banana circles.
	Ensure sustainable consumption and production patterns	The project aims to stimulate the production of handicrafts in the region, offering technical assistance and production guidance, especially to women, in partnership with a Brazilian visual artist. Products from the region will be marketed through the project website
	Take urgent action to combat climate change and its impacts	Protection of the forest area. Annual training of the community in firefighting. Training of forest patrols with members of local communities. With this, the project involves the avoidance of conversion 58,504 hectares of forest over its lifespan and it is an essential activity for climate change mitigation.
	Protect, restore, and promote the sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, halt and reverse land degradation and halt the loss of biodiversity	Given the trend towards cattle ranching in the region, the project will offer technical assistance in sustainable pasture management practices to local producers. These measures should avoid the need to open new pasture areas, improve productivity, lower the costs of raising animals and mitigate soil degradation.

1.18 Additional Information Relevant to the Project

Leakage Management

Avoided unplanned deforestation component

The AUD leakage assessment for this project on AUD component were determined as per VMD0010 – “Estimation of emissions from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation (LK-ASU)”, which sets out a procedure for assessing

and estimating the increase in GHG emissions resulting from unplanned deforestation displacement from the project boundary. Regardless of accounting of GHG emissions from possible unplanned deforestation displacement, the IACO REDD+ project addressed several leakage management activities, which were designed based on the social diagnosis carried out in the project area and in neighbor communities. As the unplanned deforestation is likely to occur in “transition configuration” on IACO REDD+, two different groups of deforestation agents may be displaced: Local Deforestation Agents and immigrant Deforestation Agents.

The following leakage activities were designed to prevent deforestation of local agents:

- Training courses. Training will be offered on different topics, but especially on environmental awareness and climate change, biodiversity and ecosystem services and sustainable management of forests, agricultural lands, and grasslands. These trainings will be open to anyone who is interested and will be disseminated to the entire local community and neighbors to the project area.
- Improvement and diversification of the local economy. the IACO REDD+ project will provide technical assistance to support existing agricultural activities and promoting culture by encouraging handicrafts for women residents in project area region. Project proponents are committed to complete transparency and will not benefit financially from sales.
- Combating illegal land occupation. Illegal land occupation and logging will be tackled with support from the local community and the project's proximity to stakeholders.

As can be difficult to reach and influence the land use decisions of immigrant agents of deforestation, the main leakage management activity will be to bring people involved with the project closer to land occupation laws and institutions that can act promptly to prevent invasions. Thus, the following leakage activity was designed to prevent deforestation of immigrant agents:

- Legal support focused on land issues. The IACO REDD+ Project proponents will provide legal support to residents of the project area zone. In addition to the land tenure regularization intention, information about the land laws in force in the State of Acre and in Brazil will be presented.

Avoided planned deforestation component

As determined as per VMD009 “Estimation of emissions from activity shifting for avoiding planned deforestation/forest degradation and avoiding planned wetland degradation (LK-ASP)” by planned deforestation, the leakage will be managed by monitoring conversion events from other private areas of the same IACO REDD+ landowners. The geographical coordinates for the areas that will be monitored as part of the leakage management plan will be available in validation.

Commercially Sensitive Information

There is no commercially sensitive information omitted from the public Project Description by the project proponents.

Further Information

There is no further relevant information to be included about the project.

2 SAFEGUARDS

2.1 No Net Harm

2.2 Local Stakeholder Consultation

2.3 Environmental Impact

2.4 Public Comments

2.5 AFOLU-Specific Safeguards

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The project activity will apply the approved VCS Methodology VM0007 “REDD+ Methodology Framework (REDD+ MF)”, Version 1.6, 08 September 2020.

Furthermore, the following modules and tools were used:

- VMD0001 Estimation of carbon stocks in the above- and belowground biomass in the live tree and non-tree pools (CP-AB), Version v1.1, 11 October 2013.
- VMD0005 *Estimation of carbon stocks in the long-term wood products pool (CP-W), Version 1.1, 20 November 2012.*
- VDM0006 Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest degradation and planned wetland degradation (BL-PL), Version 1.3, 08 September 2020.
- VMD0007 Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation (BL-UP), Version v3.3, 08 September 2020.
- VMD0009 Estimation of emissions from activity shifting for avoiding planned deforestation/forest degradation and avoiding planned wetland degradation (LK-ASP)
- VMD0010 Estimation of emissions from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation (LK-ASU), Version 1.2, 08 September 2020.
- VMD0011 Estimation of emissions from market-effects (LK-ME), version 1.1, 03 December 2010.
- VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E-BPB), version 2.2, 08 September 2020.
- VMD0015 Methods for monitoring of greenhouse gas emissions and removals in REDD project activities (M-REDD), Version 2.2, 08 September 2020.

- VMD0016 Methods for stratification of the project area (X-STR), Version 1.2, 08 September 2020
- VMD0017 Estimation of uncertainty for REDD+ project activities (X-UNC), Version 2.2, 08 September 2020.
- VT0001 Tool for the demonstration and assessment of additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) project activities (T-ADD), Version 3.0, 1 February 2012
- VCS AFOLU Non-Permanence Risk Tool (T-BAR), Version 4.0, 19 September 2019

3.2 Applicability of Methodology

Table 3 - Applicability conditions and project adherence for the REDD methodology framework module.

Module	Applicability condition	Project adherence
VM0007 - REDD-MF – Section 4.2.1	Land in the project area has qualified as forest (following the definition used by VCS) at least 10 years before the project start date.	Condition satisfied. Satellite images (Figure 1 – Forest cover of IACO REDD+ Project area in 2012 and 2020.Fonte: MapBiomass Collection 6.0) show that 100% of the project area is qualified as forest in 2011, that is, at least 10 years before the project start date.
	If land within the project area is peatland and emissions from the soil carbon pool are deemed significant, the relevant WRC modules (see Table 1) must be applied alongside other relevant modules.	Not applicable. The land within the project area is not classified as peatland.
	Baseline deforestation and forest degradation in the project area fall within one or more of the following categories: • Unplanned deforestation (VCS category AUDD); • Planned deforestation/degradation (VCS category APD);	Condition satisfied. As described in section 3.4, the project area presents an imminent risk of deforestation by agents who do not have legal and incontestable documents on the right to deforest and are residents of nearby regions or immigrants. Thus, the baseline deforestation falls into the category of unplanned deforestation (AUD). In addition, as all properties located in the IACO REDD+ project area are in accordance with the Brazilian Forest Law that allows the landowner to legally suppress the forest areas, the conversion of forest lands to

		deforested conditions is legally permitted and considered a risk. Thus, this baseline deforestation falls into the category of planned deforestation (APD).
	Leakage avoidance activities must not include: • Agricultural lands that are flooded to increase production (e.g, paddy rice); • Intensifying livestock production through the use of feed-lots and/or manure lagoons.	Condition satisfied. Neither establishment of agriculture on flooded land, nor intensification of livestock production through use of “feed-lots” and/or manure lagoons is promoted by the project.
VM0007 - REDD-MF – Section 4.2.2	Baseline agents of deforestation must: (i) clear the land for settlements, crop production (agriculturalist) or ranching, where such clearing for crop production or ranching does not amount to large scale industrial agriculture activities; (ii) have no documented and uncontested legal right to deforest the land for these purposes; and (iii) be either residents in the Reference Region for Deforestation or immigrants. Under any other condition this methodology must not be used.	Conditions satisfied. As shown in section 3.4, agents of deforestation clear the land for cattle ranching or subsistence colonization, have no documented and uncontested legal right, and are either residents in the Reference Region for Deforestation or immigrants.
	If, in the baseline scenario of avoiding unplanned deforestation project activities, post-deforestation land use constitutes reforestation, this methodology may not be used.	Condition satisfied. The post-deforestation baseline scenario does not include reforestation as alternative land use.
VMD0001 – CP-AB	This module is applicable to all forest types and age classes. Inclusion of the aboveground tree biomass pool as part of the project boundary is mandatory as per the framework module REDD-MF.	Mandatory module. The inclusion of the aboveground tree biomass reservoir as part of the project boundary is mandatory for all AFOLU projects. In addition, this module will be used to calculate the aboveground biomass of non-tree vegetation, which in the project region is significant given the dominance of bamboo and palm trees. All applicability conditions were met.

VDM0005-CP-W	This module is applicable to all cases where wood is harvested for conversion to wood products for commercial markets, for all forest types and age classes.	Mandatory module since logging operations are expected to happen in the baseline scenario prior to the conversion of forest to non-forest.
VMD0006 – BL-PL	The module is applicable for estimating the baseline emissions on forest lands (usually privately or government owned) that are legally authorized and documented to be converted to non-forest land.	Condition satisfied. The module is mandatory for all projects where the baseline deforestation falls into the category of planned deforestation (APD). As the IACO REDD+ project is in full compliance with the forest code (see Section 1.14), any forest conversion should be formally authorized by the competent body.
VMD0007 – BL-UP	The module shall be applied to all project activities where the baseline agents of deforestation: (i) clear the land for settlements, crop production (agriculturalist) or ranching, where such clearing for crop production or ranching does not amount to large scale industrial agriculture activities; (ii) have no documented and uncontested legal right to deforest the land for these purposes; and (iii) are either resident in the region (reference region—cf. section 1 below) or immigrants.	Condition satisfied. The module is mandatory for all projects that the baseline deforestation falls into the category of unplanned deforestation (AUD). As the IACO REDD+ area has no income, landowners cannot afford the efforts and costs to maintain long-term surveillance of the project boundaries to avoid encroachment on the areas.
	Where pre-project, unsustainable fuelwood collection is occurring within the project boundaries, Modules BL-DFW and LK-DFW must be used to determine potential leakage	Not applicable. Unsustainable fuelwood collection is not occurring within the REDD project boundaries. In interviews with local residents (see section 2.5), no family claimed to use fuelwood.
VMD0009 – LK-ASP	The module is applicable for estimating the leakage emissions due to activity shifting from forest lands that are legally authorized and documented to be converted to non-forest land, including activity shifting to forested wetland that is	Mandatory module. The IACO REDD+ Project met the applicability condition of this module because the baseline scenario is the conversion of forest lands that are legally authorized and

	drained or degraded as a consequence of project implementation.	documented to non-forest land. This module is used to estimate the le
VMD0010 – LK-ASU	Activities subject to potential displacement are conversion of forest land to grazing lands, crop lands, and other land uses.	As the project baseline is unplanned deforestation with the conversion of forest land to grazing lands or croplands, this module is mandatory.
VMD0011- LK-ME	This module is applicable for calculating market-effects leakage from REDD projects that are anticipated to reduce levels of wood harvest substantially and permanently.	As the project baseline has wood sales within the APD component, this module is mandatory.
VMD0013 – E-BPB	This module is applicable to REDD project activities with emissions from biomass burning and REDD-WRC project activities with emissions from biomass and/or peat burning. This module is also applicable to RWE and ARR-RWE project activities with emissions from peat burning.	As it is common for deforestation to occur by burning and cutting ³⁵ in the project region, this module is mandatory.
VMD0015 – M-REDD	The module is mandatory for REDD, CIW-REDD, RWE-REDD and stand-alone CIW project activities	Mandatory module. The IACO REDD+ is a REDD project.
	Where selective logging is taking place in the project case: - Emissions from logging may be omitted if it can be demonstrated the emissions are de minimis using Tool T-SIG. - If emissions from logging are not omitted as de minimis, logging may only take place within forest management areas that possess and maintain a Forest Stewardship Council (FSC) certificate for the years when the selective logging occurs. - Logging operations may only conduct selective logging that maintains a	Not applicable. The project does not involve selective logging.

³⁵ Branco-Acre, Rio. "DINÂMICA DO DESMATAMENTO EM 2017 NO ESTADO DO ACRE-PRODES." Available at: < http://semapi.acre.gov.br/wp-content/uploads/sites/20/2020/06/Relatorio-Desmatamento-PRODES-2017-FINAL_20181015_Revisado_20200413.pdf >. Last access: 29th April, 2022

	land cover that meets the definition of forest within the project boundary. • All trees cut for timber extraction during logging operations must have a DBH greater than 30 cm. • During logging operations, only the bole/log of the felled tree may be removed. The top/crown of the tree must remain within the forested area. • The logging practices cannot include the piling and/or burning of logging slash • Volume of timber harvested must be measured and monitored.	
VMD0016 – X-STR	Any module referencing strata i must be used in combination with this module. In case of REDD, above-ground biomass stratification is only used for pre-deforestation forest classes, and strata are the same in the baseline and the project scenario. Post deforestation land uses are not stratified. Instead, average post deforestation stock values (e.g. “Simple” or “Historical area-weighted” approaches are used, as per Module BL-UP).	Mandatory module. As described in section 3.4, different forest types are existing in the project area characterizing different strata.
	For peatland rewetting and conservation project activities this module must be used to delineate non peat versus peat and to stratify the peat according to peat depth and soil emission characteristics, unless it can be demonstrated that the expected emissions from the soil organic carbon pool or change in the soil organic carbon pool in the project scenario is de minimis, In the case of WRC project activities, the project boundary must be designed such that the negative effect of drainage activities that occur outside the project area on the project GHG benefits are minimized	Not applicable. The land within the project area is not classified as peatland or wetland.

VMD0017 – X-UNC	This module is mandatory when using methodology REDD-MF.	Mandatory module. The IACO REDD is a REDD project.
T-ADD	Forestation of the land within the proposed project boundary performed with or without being registered as the A/R CDM project activity shall not lead to violation of any applicable law even if the law is not enforced.	Condition satisfied. Project activities do not violate any applicable laws. They all aligned with government guidelines.
	This tool is not applicable to small - scale afforestation and reforestation project activities	Condition satisfied. The IACO REDD+ is a REDD project.
T-BAR		Mandatory tool for AFOLU projects.

3.3 Project Boundary

Geographical boundaries

According to the VM0007 methodology, where multiple baselines exist (e.g., planned deforestation and unplanned deforestation) there must be no overlap in boundaries between areas appropriate to each of the baselines. Thus, the two project types cannot occur on the same piece of land. Therefore, geographical boundaries were defined separately for unplanned deforestation activities, which cover 117,690.16 hectares, and for planned deforestation activities, which cover 29.708,23 hectares or 20% of the total property area.

Avoided unplanned deforestation component

According to the VM0007 methodology, project activity to avoided unplanned deforestation is conducted in the spatial delineation of Project Area, Reference Region, and Leakage Belt following the module BL-UP. The project area under avoided unplanned deforestation activities is constituted of 117,690.16 hectares which corresponds to the forest that has been present for more than 10 years before the project start date (**Figure 13**). The definition of the Reference Region and the Leakage Belt was performed based on a detailed analysis of the factors to be characterized (such as agents of deforestation) or quantified (such as the proportion of forest types).

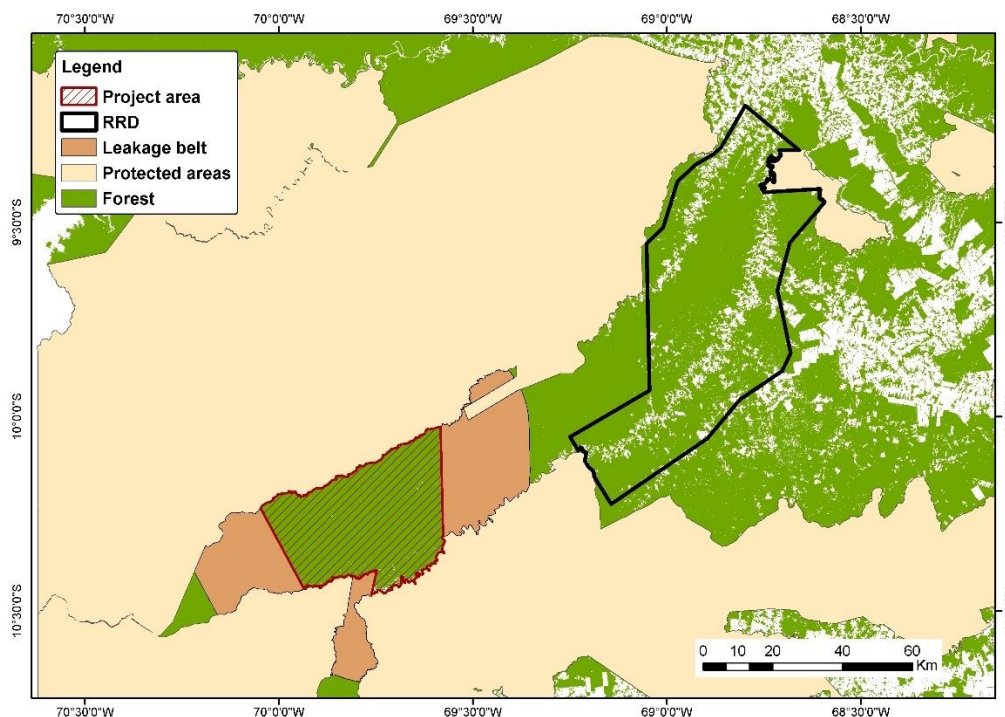


Figure 13– Localization of project area, leakage belt and RRD.

Although the reference region is composed of a Reference region for projecting the rate of deforestation (RRD) and a Reference region for projecting the location of deforestation (RRL), the RRD was the only Reference Region defined in this project. RRL is not required in the case of a transition configuration of deforestation, where it can be shown that $\geq 25\%$ of the project geographic boundary is within 50 m of land that has been anthropogenically deforested within the 10 years before the project start date (see Figure 3).

The RRD was defined based on the characteristics of deforestation agents (access and mobility), laws, statutes, regulatory frameworks, social customs, and environmental conditions. Thus, the RRD was defined within the same municipality and state, to ensure that the systems of laws and governance, social structure, and customs are similar to those occurring in the project area. All areas that have restricted access and different governance systems, such as conservation units and indigenous lands) were excluded from the reference region. Moreover, the main mobility vector for deforestation agents are the same navigable rivers. Finally, the type of vegetation and soil, slope, land cover, and use, precipitation, and temperature were similar to the conditions found in the project area (see **Table 4**).

The minimum area of the RRD was defined by following equations specified in BL-UP:

$$\text{MREF} = \text{RAF} * \text{PA}$$

$$\text{RAF} = 7500 * \text{PA}^{-0.7},$$

Where:

MREF is Minimum size of reference region for projecting rate of deforestation; ha

PA is Unplanned deforestation project area; ha

RAF is Reference Area Factor. Factor to multiply times project area to get minimum reference area; dimensionless

Given the AUD project area have in 117,690.16 ha, the RRD should be at least 249,048.14 ha. The selected RRD had 341.533,38 ha of forest cover at the beginning of the reference period (in 2012), which meets the minimum required by the methodology **Figure 14**.

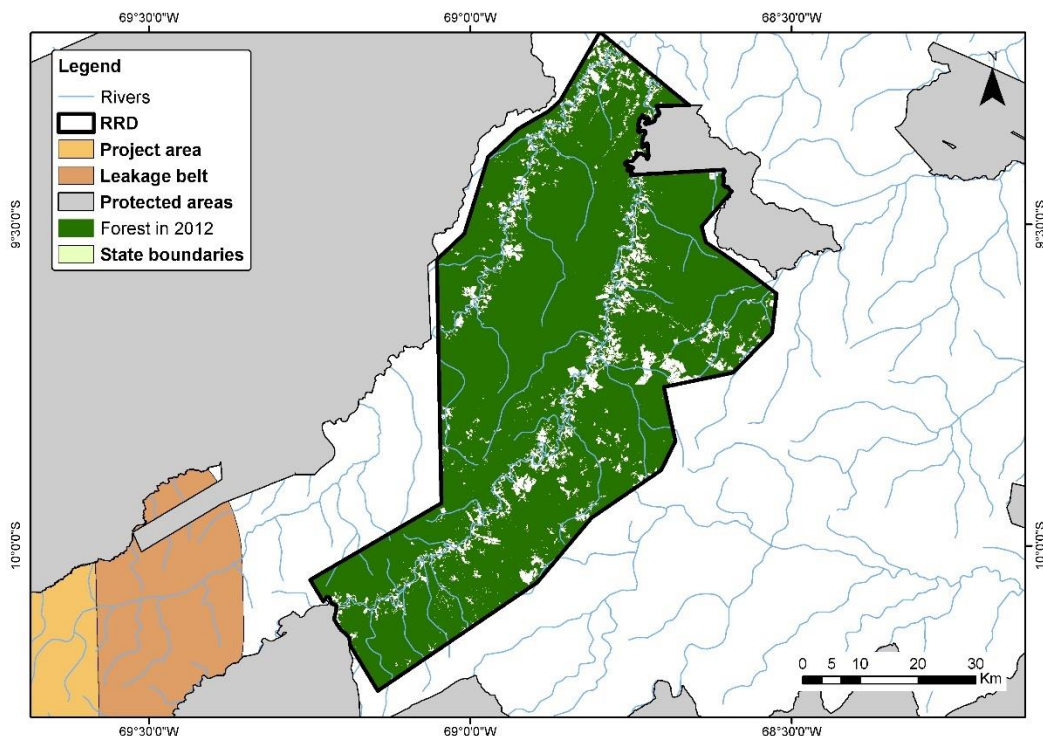


Figure 14 – Forest cover's RRD in 2012.

The Leakage Belt is the area surrounding or near the project area where leakage caused by activity displacement is expected to occur. It was defined by the distribution of closed forests in the region and the methodological requisite of similar landscape conditions and accessibility of deforestation agents with project area and a minimum area requirement (i.e., $\geq 90\%$ of the project area). The leakage belt covers 154.409,6 hectares of forest which is more than 100% of the project area.

Table 4 - Comparison of vegetation, elevation, slope, and soil type of the project area, reference region for deforestation (RRD) and leakage belt.

	Category	Project area	RRD	Leakage belt
Vegetation (%)	Alluvial open rainforest	5.48	10.61	8.05
	Lowland open rainforest	94.52	81.39	91.94

	Category	Project area	RRD	Leakage belt
	Other physiognomies	0	8.00	0.01
Elevation (%)	0 – 500 m	100	100	100
Slope (%)	≤ 15%	100	99.98	100
Soil (%)	> 15%	0	0.02	0
	Gleisol	5.57	14.18	8.82
	Luvisol	63.86	19.55	52.75
	Argisol	30.57	66.26	38.42
	Other (water courses)	0	0.01	0
Rivers (m/km ²)		691	765	672

Avoided planned deforestation component

The APD project area is constituted of 29.708,23 hectares which corresponds to the forest that can be legally converted to other land uses for commercial activities, with the authorization of the responsible environmental agency in each state.

Temporal boundaries

Avoided unplanned and planned deforestation component

Start date and end date of the historical reference period

According to VM0007, the starting date of the historical reference period must be between 9 and 12 years in the past and the end date must be within two years before project start date. Since the project start date is defined in 2022, the reference period for calculating the rate of unplanned and planned deforestation was defined as the period between 2011 and 2020.

Carbon pools

The IACO REDD+ Project considers the carbon pools listed in the tables below. Their inclusion or exclusion within the boundary of the project activity, as well as the respective justification/explanation, are also described in the tables below.

Avoided unplanned deforestation component

The carbon stock assessment was designed by the protocol established in the module VMD0007 (BL-UP) and VMD0001 (CP-AB).

Carbon pools	Included?	Justification/Explanation
Above-ground of live trees	Yes	This pool is mandatory and always significant
Above-ground live of non-trees	Yes	The inclusion of this component is particularly important in the project area since most of the species present in the area are bamboos and palms.
Below-ground of live trees	Yes	Significant carbon pool.
Below-ground of live non-trees	Yes	The inclusion of this component is particularly important in the project area since most of the species present in the area are bamboos and palms.
Deadwood	No	Conservatively excluded, as allowed by methodology.
Harvested wood products	No	No long-term wood product activities were observed in the project area. According to the socio-environmental diagnosis made with the community that lives around the project area (see section 2), there is no commercialization of wood products.
Litter	No	Conservatively excluded, as allowed by methodology.
Soil organic carbon	No	Conservatively excluded, as allowed by methodology.

GHG emission sources included in the IACO REDD+ project boundary are listed in the table below with the respective justification/explanation for inclusion or exclusion.

Table 5. Sources and GHG included or excluded within the boundary of the proposed AUD project activity

Source		Gas	Included?	Justification/Explanation
Baseline	Biomass burning	CO ₂	No	Excluded as recommended by the applied methodology. Counted as carbon stock change.
		CH ₄	Yes	Included as non-CO2 emissions from biomass burning in the baseline scenario, according to the methodology.
		N ₂ O	Yes	Included as non-CO2 emissions from biomass burning in the baseline scenario, according to the methodology.
		Other	No	No other GHG gases were considered in the baseline scenario.

Source		Gas	Included?	Justification/Explanation
Project	Livestock emissions	CO ₂	No	Not a significant source
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
		Other	No	No other GHG gases were considered in the baseline scenario
	Biomass burning	CO ₂	No	Excluded as recommended by the applied methodology. Counted as carbon stock change.
		CH ₄	Yes	Included as non-CO ₂ emissions from biomass burning in the project scenario, according to the methodology.
		N ₂ O	Yes	Included as non-CO ₂ emissions from biomass burning in the project scenario, according to the methodology.
		Other	No	No other GHG gases were considered in this project activity.
	Livestock emissions	CO ₂	No	Not a significant source
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
		Other	No	No other GHG gases were considered in the baseline scenario

Avoided planned deforestation component

The carbon stock assessment was designed by the protocol established in the module VMD0006 (BL-PL) and VMD0001 (CP-AB).

Carbon pools	Included?	Justification/Explanation
Above-ground of live trees	Yes	This pool is mandatory and always significant
Above-ground live of non-trees	Yes	The inclusion of this component is particularly important in the project area since most of the species present in the area are bamboos and palms.
Below-ground of live trees	Yes	Significant carbon pool.

Carbon pools	Included?	Justification/Explanation
Below-ground of live non-trees	Yes	The inclusion of this component is particularly important in the project area since most of the species present in the area are bamboos and palms.
Deadwood	No	Conservatively excluded, as allowed by methodology.
Harvested wood products	Yes	Inclusion of the harvested wood pool as part of the project boundary is mandatory when the process of the deforestation involves timber harvesting for commercial markets.
Litter	No	Conservatively excluded, as allowed by methodology.
Soil organic carbon	No	Conservatively excluded, as allowed by methodology.

Table 6. Sources and GHG included or excluded within the boundary of the proposed APD project activity

Source		Gas	Included?	Justification/Explanation
Baseline	Biomass burning	CO ₂	No	Excluded as recommended by the applied methodology. Counted as carbon stock change.
		CH ₄	Yes	Included as non-CO2 emissions from biomass burning in the baseline scenario, according to the methodology.
		N ₂ O	Yes	Included as non-CO2 emissions from biomass burning in the baseline scenario, according to the methodology.
		Other	No	No other GHG gases were considered in the baseline scenario.
	Combustion of Fossil Fuels	CO ₂	No	Carbon emissions from harvesting equipment, log transport in the baseline scenario can be conservatively excluded according to VM0007
		CH ₄	No	Non-CO2 emissions from harvesting equipment, log transport in the baseline scenario can be conservatively excluded according to VM0007
		N ₂ O	No	Non-CO2 emissions from harvesting equipment, log transport in the baseline scenario can be conservatively excluded according to VM0007

Source		Gas	Included?	Justification/Explanation
Project	Use of Fertilizer	CO ₂	No	No increase in fertilizer use is contemplated in the project case as part of leakage mitigation or any other activity.
		CH ₄	No	No increase in fertilizer use is contemplated in the project case as part of leakage mitigation or any other activity.
		N ₂ O	No	No increase in fertilizer use is contemplated in the project case as part of leakage mitigation or any other activity.
		Other	No	No increase in fertilizer use is contemplated in the project case as part of leakage mitigation or any other activity.
	Biomass burning	CO ₂	No	Excluded as recommended by the applied methodology. Counted as carbon stock change.
		CH ₄	Yes	Included as non-CO2 emissions from biomass burning in the project scenario, according to the methodology.
		N ₂ O	Yes	Included as non-CO2 emissions from biomass burning in the project scenario, according to the methodology.
		Other	No	No other GHG gases were considered in this project activity.
	Combustion of Fossil Fuels	CO ₂	No	Carbon emissions from harvesting equipment, log transport in the baseline scenario can be conservatively excluded according to VM0007
		CH ₄	No	Non-CO2 emissions from harvesting equipment, log transport in the baseline scenario can be conservatively excluded according to VM0007
		N ₂ O	No	Non-CO2 emissions from harvesting equipment, log transport in the baseline scenario can be conservatively excluded according to VM0007
	Use of Fertilizer	CO ₂	No	No increase in fertilizer use is contemplated in the project case as part of leakage mitigation or any other activity.

Source		Gas	Included?	Justification/Explanation
		CH ₄	No	No increase in fertilizer use is contemplated in the project case as part of leakage mitigation or any other activity.
		N ₂ O	No	No increase in fertilizer use is contemplated in the project case as part of leakage mitigation or any other activity.
		Other	No	No increase in fertilizer use is contemplated in the project case as part of leakage mitigation or any other activity.

3.4 Baseline Scenario

3.5 Additionality

3.6 Methodology Deviations

No methodology deviations were applied.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

4.2 Project Emissions

4.3 Leakage

4.4 Net GHG Emission Reductions and Removals

5 MONITORING

5.1 Data and Parameters Available at Validation

5.2 Monitoring Plan

APPENDIX